PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)
100-ft Ring Road, Bengaluru – 560 085, Karnataka, India

Capstone Project Report Phase-2
on

The Indian Heritage

Submitted by

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Feb 2026 – May 2026
under the guidance of

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FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER APPLICATIONS
PROGRAM – MASTER OF COMPUTER APPLICATIONS

CERTIFICATE

This is to certify that the project entitled

The Indian Heritage

is a bonafide work carried out by

Satyapriya Sahoo - PES1PG24CA154

in partial fulfillment for the completion of Capstone Project Phase-2 work in the Program of Study MCA under rules and regulations of PES University, Bengaluru during the period Feb 2026 – May 2026. The project report has been approved as it satisfies the academic requirements of 4th semester MCA.

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DECLARATION

I, **Satyapriya Sahoo,** bearing **PES1PG24CA154** hereby declare that the Capston project Phase-2 entitled, ***The Indian Heritage,*** is an original work done by me under the guidance of **Ms. Samyukta D Kumta,** *Designation*, PES University, and is being submitted in partial fulfillment of the requirements for completion of 4^{th} Semester course in the Program of Study **MCA**. All corrections/suggestions indicated for internal assessment have been incorporated in the report.

The plagiarism check has been done for the report and is below the given threshold.

I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other course.

PLACE: Bengaluru

DATE:

Satyapriya Sahoo

PES1PG24CA154

ACKNOWLEDGEMENT

I take great pleasure in expressing my sincere gratitude to all those who have guided me and supported me to successfully complete the project.

I would like to express my sincere gratitude to the Vice chancellor of PES University **Dr.J Suryaprasad** and Chairperson **Dr.Veena S** who gave me an opportunity to do the project as part of my curriculum.

I am grateful to my guide, **Ms. Samyukta D Kumta**, Assistant Professor, Department of Computer Applications, who has been my source of inspiration and providing me with guidance, encouragement and support, during the course of the project.

Satyapriya Sahoo
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ABSTRACT

The main objective of this project is to design and implement an innovative e-commerce website where customers can navigate and search for the required products and make payments for their purchases using a convenient and efficient interface. In this website, customers can search for products by browsing through the various categories available, read about the specifications of the product, put it into their cart, and then make payments for the selected item using payment options provided in the system. To achieve maximum efficiency and effectiveness, the system is designed using advanced web technologies with an organized architecture to facilitate seamless communication between its front-end and back-end layers. Apart from offering basic functionalities associated with e-commerce, there are some intelligent techniques used to make this possible with better efficiency and ease of use for users. The system has been developed keeping in view the needs of fast-loading, navigation, and responsiveness so that users can easily access the system from any location. In addition to being an effort of developing a system to perform online shopping activities, the most important aspect of this project is to offer the users a satisfactory experience. This project aims at improving the efficiency and effectiveness of the system while making it accessible to all kinds of users.

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Introduction

CHAPTER 1

Introduction

1.1 Project Description

For instance, in recent years, people have incorporated online shopping into their lives, where they gain access to goods and services regardless of their location. Nonetheless, even though numerous people use e-commerce websites, some e-commerce applications still exhibit certain constraints, including poor user interface, inefficiency, and inadequate system design. In addition, most customers find it difficult to navigate the product listing, search for items, and complete transactions on e-commerce sites because of the inadequacies mentioned above. The problems get worse when the user base and product portfolio grow larger.

From the system point of view, traditional e-commerce sites depend on primitive designs that concentrate mainly on the site's features instead of its user interface. This situation may cause some problems such as high response time, failure to personalize products for particular users, ineffective database management, and inconsistency in the system. Furthermore, a website that is not designed for expansion cannot accommodate more users, thus becoming inconsistent in its operations.

Therefore, there is an urgent need for designing an e-commerce website that incorporates essential functions while ensuring usability and consistent interactions between the user and the web application. An ideal e-commerce site should be scalable, which implies that the system will expand based on the needs of the user base.

1.2 Problem Statement

Currently, users of existing systems encounter problems in accessing and navigating through products in the most efficient manner possible. Most of the existing applications allow users to browse for products and place orders. However, such features are not enough since there is a need for an application with better performance capabilities, an intuitive interface, and a streamlined interaction process.

In some cases, the time consumed while finding products, comparing available options, or making payments is higher than it should be because of the current state of affairs. The user interface and overall performance of the system are not optimal either, which further worsens the situation. Administrators may have trouble organizing the information related to products, users, and orders, which might lead to other problems in the future.

Such problems should be resolved to ensure maximum efficiency.

1.3 Proposed Solution

As a proposed solution, the main focus will be put on the development of a scalable and effective e-commerce platform, which would solve identified limitations of existing systems. The developed system will include all elements necessary for creating a user-friendly interface and an efficient back-end system.

The system will allow browsing through product catalogues across multiple categories, obtaining details about each individual product, adding products to carts, and completing transactions. To achieve uniformity and performance, it was decided to implement a full-stack architecture for this platform, where both front-end and back-end processes will be handled within the same framework. This will make communication processes within the system much faster and more responsive.

In addition to implementing core system functionalities, the use of advanced mechanisms, such as advanced searches and fast data processing, will also be included to ensure better system performance and user experience.

1.4 Purpose

One of the primary objectives in this project is to create a convenient and effective online shopping platform that allows customers to do their transactions without compromising the performance of the system. One of the major benefits of the project is that the system can make transactions much easier for the users through a well-managed and systematic way of completing certain tasks.

Another important objective of this project is to provide a good level of convenience and user-friendly interface while performing some activities like navigating on the platform. In addition, the system will provide efficient ways of managing product details and order information.

1.5 Scope

This project's scope pertains to the design of an organized online platform to provide essential functions in the context of online retailing. The system has provisions for browsing products, navigating based on categories, searching for specific items, managing the cart, and processing orders.

The application caters to small-to-medium scale applications, ensuring seamless interaction between the user and the software, with consistent performance at all times. The scope does not entail developing complex systems that can handle intricate enterprise-level operations.

The system lacks provisions for incorporating advanced functionalities such as integration with multiple vendors, comprehensive analysis tools, and sophisticated recommendation engines. Nevertheless, the system is modular in nature, and any additions in future can be incorporated seamlessly.

Literature Survey

CHAPTER 2

LITERATURE SURVEY

2.1 RELATED WORK:-

2.1.1 Neural Information Retrieval for E-Commerce:-

Metadata:-
B. Mitra and N. Craswell, "Neural Information Retrieval for E-Commerce," 2018.

Summary:-
This study discusses the application of neural network techniques to enhance information retrieval in e-commerce platforms. The research mainly focuses on interpreting user search queries and identifying the most relevant products through deep learning methods. It also explains how semantic understanding can improve traditional keyword-based product searches.

Implementation:-
The concepts presented in this research contribute to enhancing the search functionality of The Indian Heritage platform. By utilizing intelligent search approaches, the system can provide more accurate product results based on user intent instead of relying only on exact keyword matching.

2.1.2 Enhanced Product Searching Using NLP Approaches:-

Metadata:-
SIGIR Workshop of ACM, "Improving Product Search with NLP Techniques," 2020

Summary:-
This research paper highlights the role of NLP techniques in helping out with product searches in e-commerce websites. This technique is based on analyzing the user's queries and improving the efficiency of search by comprehending its meaning.

Implementation:-
The implementation of these concepts in The Indian Heritage system will help improve the efficiency of searching for products in the system.

2.1.3 Semantic Search Using Deep Learning:-

Metadata:-
IEEE, "Semantic Search Using Deep Learning," 2021.

Summary:-
The research presents semantic search models, which utilize deep learning approaches to interpret the context behind user queries. It allows the system to go beyond matching keywords and give more meaningful outputs.

Implementation:-

The technology is employed in The Indian Heritage system to improve the accuracy of search through the identification of the intention behind the query, thus enabling users to find products that match their purpose without having to type exact keywords.

2.1.4 Deep Learning for Image-Based Product Search:-

Metadata:-

J. Johnson et al., “Deep Learning for Image-Based Product Search,” 2016.

Summary:-

This article describes the process of using deep learning algorithms to create an image-based product search engine. The method involves extracting visual features and comparing them to match similar products.

Implementation:-

This innovation contributes to the visual search feature in The Indian Heritage system, where users can find products through images, leading to better product search experiences.

2.1.5 Visual Search for E-Commerce using CNNs:-

Metadata:-

IEEE, "Visual Search for E-Commerce Using CNNs", 2020.

Summary:-

This study is centered around the application of Convolutional Neural Network models in e-commerce systems. It increases the effectiveness of visual product search.

Implementation:-

The methods outlined above improve the functionality of the image searching tool used within The Indian Heritage project to increase efficiency.

2.1.6 Development of Scalable E-Commerce Systems:-

Metadata:-

IEEE, "Design of Scalable E-Commerce Systems", 2018.

Summary:-

This article highlights several essential factors needed for the design of scalable e-commerce systems.

Implementation:-

These factors are considered in the design of The Indian Heritage system.

2.1.7 Microservices Architecture vs Monolithic Architecture

Metadata:-
N. Dragoni et al., "Microservices vs Monolithic Architecture," 2017.

Summary:-
In this paper, an investigation of the pros and cons of both microservice and monolithic architecture systems is provided.

Implementation:-
The Indian Heritage application utilizes a unified architecture system; therefore, this research paper will aid in supporting the implementation rationale behind it while explaining how the system can remain efficient and simple despite using it.

2.1.8 Security Problems with Online Payments

Metadata:-
S. Kaur & R.K. Gupta, "Security Issues in Online Payment Systems," 2019.

Summary:-
The document discusses some of the main security problems encountered during online payments, such as fraud and theft of customer's personal information, and mentions the importance of authentication methods in online payment services.

Implementation:-
These ideas will help in implementing proper online payment services in the application.

2.1.9 Growth of E-Commerce in India:-

Metadata:-
P. R. Singh & A. Mehta, “Growth of E-Commerce in India,” 2020.

Summary:-
This article discusses the growth of e-commerce in India and how users adopt e-commerce in their daily lives and business operations.

Implementation:-
The information gathered from this paper will be used in creating The Indian Heritage platform. This platform will emphasize Indian products in the online marketplace.

2.1.10 Digital Payment Systems in India:-

Metadata:-

IEEE, "Digital Payment Systems in India," 2020.

Summary:-

This paper talks about the growth of digital payments in India and how they affect user trust in online transactions.

Implementation:-

This paper will help implement the feature of digital payment into The Indian Heritage platform to ensure a seamless transaction process.

2.2 Existing System:-

Platform	What It Offers	Why It's Useful	Where It Falls Short	Overall
Flipkart	Online platform for product browsing, payments, and delivery	Used extensively in India; offers wide variety of products; provides seamless user experience	Limited in-depth personalization; not tailored for specific domains	Very high
Messo	Platform for resale of products; social commerce	Facilitates product selling; low cost of entry; good for small business ventures	Requires initial configuration; many features have charges; depends on third-party add-ons	Moderate to high
Shopify	Online store platform that allows custom design and management	Highly flexible; supports scalable businesses; allows customization; includes effective store management tools	Requires initial configuration; many features have charges; depends on third-party add-ons	Very high
Google Search-Based Shopping	System that allows product discovery using search engines and external websites	Fast product discovery; wide reach; accessible to multiple sellers	Lack of personalizatio n; no defined or organized shopping experience	Moderate

2.3 FEASIBILITY STUDY:-

2.3.1 Introduction:-

The feasibility study determines the viability of implementing the proposed information system known as “The Indian Heritage” under the specified circumstances. This proposed system is a well-organized online shopping system where people can access, search, and buy goods using its user-friendly interface. In addition, it features improved functionalities like smart searching and efficient data management systems. The objective of this paper is to examine the feasibility of the information system in terms of technical, economic, and operational feasibility.

2.3.2 Technical Feasibility:-

The proposed solution is technically feasible because it makes use of advanced web technologies that ensure scalability, maintainability, and performance.

Technologies Employed:-

Frontend: React.js, Tailwind CSS
Backend/ Framework: Next.js (Full stack architecture)
Database: MongoDB
Authentication: NextAuth
Image Storage: ImageKit
AI/Search Optimization: Embedding based search algorithms
Hosting Platform: Vercel

In addition, the system adopts a full-stack approach to development where both the frontend and backend functionalities are combined in one package. This makes interactions between the two sides much simpler and enhances performance.

Moreover:-

Effective product searches are facilitated by enhanced query optimization
Data interactions are performed in real-time
User authentication is provided by secure mechanisms
Web hosting is provided by cloud-based infrastructure

Conclusion:
The proposed solution is technologically feasible as it employs dependable web technologies and architecture designs that meet current requirements and provide room for future developments.

2.3.3 Economic Feasibility:-

The suggested system is economically feasible as it involves lower development and operation costs, yet provides maximum benefits.

Costs Associated With:

Development and Implementation Effort
Cloud Deployment
System Maintenance and Upgrades
Storage of Data and Media Handling

Benefits Obtained From:

Low cost due to the application of open source technologies
Low infrastructure requirements due to cloud deployment
Optimized system structure minimizing the maintenance cost

Considering that the technology utilized is widely available, the overall cost of the solution will be relatively low. The benefits obtained from automation and effective product management will outweigh the costs involved.

Conclusion:
The system is economically feasible and will provide maximum benefits in the long run.

2.3.4 Operational Feasibility:-

This system is developed to be efficient, user-friendly, and easy to operate by the customers and the administrators.

Users Identified:

Customer
Admin

Strengths of the Operational Feasibility:-

Well-organized interface to explore the products
Effective mechanism for searching products
Efficient order and cart management feature
Secure login mechanism
Responsive system interface

The system operates in such a way that it doesn't require any technical expertise from the user. Operations related to administration such as product handling and order management are also well-structured and easy to manage.

Conclusion:
The operational feasibility of this system can be considered to be good.

Hardware and Software Requirements

CHAPTER 3

HARDWARE AND SOFTWARE REQUIREMENTS

3.1 Hardware Requirement:-

Requirements of the Client Side / End User:-

Mobile Device:-

• Android 8.0 or later & iOS 12 or later & 3GB RAM at least
• Modern Web Browser like Chrome, Safari or Edge
• Mobile Device with touch screen interface for easy navigation
• Internet Connectivity should be stable with minimum speed of 1 Mbps

Desktop/Laptop:-

• Latest Version of OS - Windows 10/macOS/Linux(Ubuntu 18.04 or later)
• Minimum RAM = 4 GB or recommended 8 GB RAM
• Web Browser – Chrome, Firefox, Safari, Edge (Latest Versions)
• Standard Display
• Internet Connectivity must have broadband connectivity with at least 5 Mbps speed

Requirements of the Development System Used for Implementation:-

Device – MacBook Air (13-inch, 2025)
Chipset – Apple M4
Memory = 16 GB
Storage = 256 GB SSD
Operating System – macOS Tahoe (Version 26.4.1)
Display – 13.6-inch Retina Display (2560 × 1664)
Internet Connectivity - Broadband Connection

Requirements of the Server Side

Production Environment:-

• Hosting Platform – Cloud-based e.g. Vercel
• Scalable resources (CPU and Memory)
• Storage provided by cloud database service like MongoDB Atlas.
• CDN for fast content & images
• Network Infrastructure

3.2 Software Requirement:-

Frontend (User Interface): TypeScript, JavaScript (React.js), HTML, CSS,

Tailwind CSS

Backend (Application Programming Interface): JavaScript (Next.js –

Full-stack framework)User Authentication: NextAuth (JWT based user authentication)

Database: MongoDB (Non-relational database)

Managing Images: ImageKit (For storing and serving images)

Searching/AI Improvements: Embedding based search to improve user experience with products

Software Used: Node.js (Version 20+), npm

Editor: Visual Studio Code

Source Control: Git

Hosting Service: Vercel (Cloud-hosting service)

Browsers Supported: Google Chrome, Mozilla Firefox, Safari, Microsoft Edge

Software Requirements Specification

CHAPTER 4

SOFTWARE REQUIREMENTS SPECIFICATION

4.1 Users:-

4.1.1 Admin:-
The administrator is responsible for managing and supervising the overall activities of the online platform. This role involves ensuring the smooth functioning of the system and monitoring all operations carried out within the application.

The admin has the authority to manage product listings, supervise seller activities, and monitor customer orders. In addition, the administrator handles important system-level operations such as maintaining data consistency, resolving issues, and ensuring that transactions are secure and properly processed. The admin can also update or remove inappropriate content and manage user accounts whenever required.

4.1.2 Seller:-
The seller is responsible for uploading and maintaining products on the platform. Sellers are allowed to add new products, modify product details, and manage stock availability according to inventory conditions.

Furthermore, sellers can access and manage orders related to their products to ensure proper processing and timely delivery. They are also able to control product pricing, descriptions, and product images to present their products effectively on the platform.

4.1.3 Buyer (Customer):-

The buyer is the end user who interacts with the platform to browse and purchase products. Buyers can explore products from different categories, search for specific items, and access detailed information about products before making purchases.

Customers can add products to the shopping cart, manage selected items, and complete transactions using the available payment methods. In addition, buyers can track placed orders, review purchase history, and update their account details. The platform is designed to provide a smooth and user-friendly shopping experience for customers.

4.1.4 System (Optional Role):-

The system acts as the central component responsible for handling communication between users and backend services. It manages various operations such as request processing, product data management, secure user authentication, and performance optimization.

Although the system is not considered a direct user, it plays an important role in maintaining smooth interaction and coordination between all components of the platform.

4.2 Functional Requirement:-

User registration and login/Authentication:-
The system facilitates users to create accounts and login using their credentials. The proper authentication process helps in protecting the system from any unauthorized activity. Maintaining user sessions helps them to access the account during several visits. The feature helps in managing personalized information including orders and account details.

Browsing products within the system:-
A user-friendly interface is provided to the users for easy navigation of various products. Products are categorized into different categories so that a user may easily move through various sections without any hassle. Thus, users are provided the best user experience while browsing through various products.

Searching Products using Keywords:-
The system offers searching functionality where users can find products using keywords. The system finds out relevant results for users based on the keywords input by users. This process saves time as well as offers better convenience to the users.

NLP-based Search (enhancing search):-
In addition to keyword searches, this system offers NLP-based search. By using NLP techniques, the query intent is analyzed by the system and it fetches the relevant result based on the query intent, instead of just searching through the input keywords. This increases the accuracy and efficiency of search functionality.

Image-Based Product Search:-
This system provides another searching functionality through which users may search for products via an image rather than typing in some keywords. The system analyzes the image properties and finds out the relevant result based on those properties. Thus, in case users do not know the keyword for a product, this functionality comes in handy.

Product Details Viewing:-
Users can view details of the products that include descriptions, prices, and even images of them. It allows them to have an insight into what product they are buying, hence aiding in their purchasing decisions. This will help prevent any kind of confusion by providing the information in a clear manner.

Cart Management: -
Users can put the selected product in the cart for purchase at a later stage. They can change the quantity of the product or delete the selected item from the cart prior to purchasing. They can check what product they added to the cart, which will be stored till the order is made.

Order Making: -
Users can make an order by adding the required product from the cart. All the required details, i.e., order-related product details and users' personal details, are stored with the system. This feature is the basic core transaction feature.

Payment Making: -
The site will provide facilities to complete transactions through payments. The payment

details of the customers will be taken in a secured manner. After the payment is made successfully, the order will be marked as completed in the system.

Order Tracking & History: -

This feature will allow the users to see the order made by them and their transaction histories. This will enable them to keep a track of their order and manage their order history.

Seller Product Management:-

Sellers can add new products, modify product details, and maintain inventories. It helps them in managing the product listings. This will facilitate the maintenance of relevant product information on the site. It makes the management of products easier.

Seller Order Management:-

Sellers can manage their orders related to the products. They can process orders, track orders, and update orders. It helps the sellers to keep track of the orders made by the customers. It ensures coordination between buyer and seller.

Admin Control & Management:-

The admin controls all activities within the platform. He is responsible for managing all the users, products, and other system issues as per the need. His task is to ensure the smooth running of the platform. It is necessary for ensuring system stability.

User Account Management:-

Users have an option of updating the details of their profile. This will help in managing the data associated with each user. It makes it easy for keeping record of transaction and interaction done through the application.

System Performance & Data Handling:-

The system provides better performance for the users. It ensures that data is processed correctly and accurately. It provides seamless communication between frontend and backend. It facilitates consistent performance regardless of any number of user operations.

4.3 Non-Functional Requirement:-

Security:-
Security is crucial for any e-commerce website because of the nature of sensitive information being handled. There are authentication mechanisms put in place to ensure that only authorized persons can access the user account. Login and payment details must be encrypted to ensure maximum security while processing transactions.

Scalability:-
A scalable application must handle many users and products without compromising its efficiency. There should be room for growth as far as the quantity of data and activity are concerned. The platform utilizes cloud computing services to facilitate easy scalability.

Availability:-
Availability of any application means that it is available for use by the target audience at all times. The system must have reliable hosts to keep it running and accessible all the time. In case of a system failure, it must be capable of recovery.

Usability:-
Usability is an important quality of any application as far as user interaction is concerned. The platform must be navigable and easy to use. User interface should allow users to interact with the system without any challenges. There should be flexibility in terms of devices to be used.

Performance:-
Performance refers to the ability of the platform to respond to user input efficiently. There must be efficient processing of data during tasks such as browsing, searching, and buying products. Users must be capable of completing tasks with minimal waiting time.

Reliability:-
This is an essential characteristic of any system because it must function consistently without any faults. The system must have efficient error handling procedures to make sure that there is no loss of data during processes.

System Design

Chapter 5

SYSTEM DESIGN

5.1 ARCHITECTURE:-

5.1.1 DIAGRAM:-

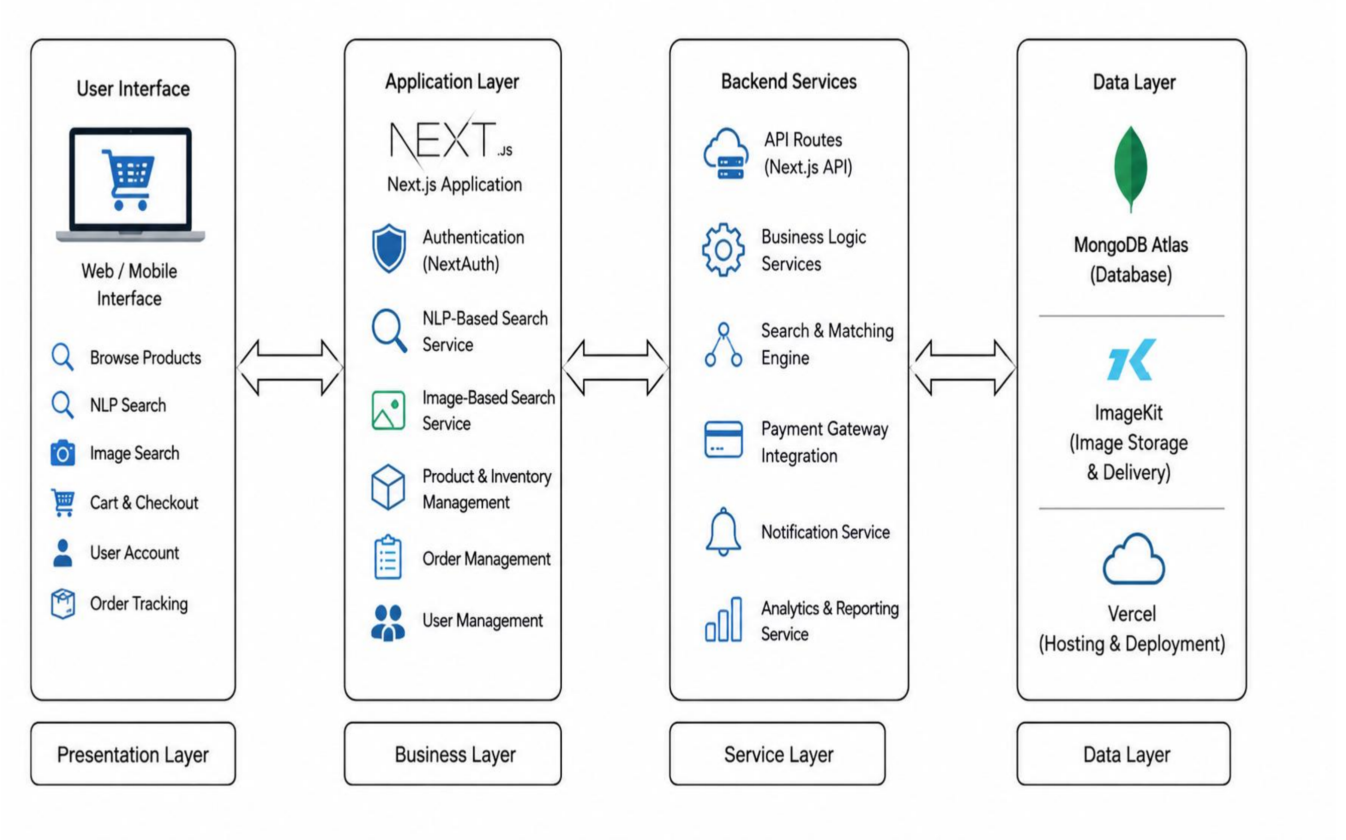

Figure 5.1.1 : Architecture Diagram

5.1.2 Explanation:-

The Indian Heritage platform is designed using a layered architecture to provide better organization and efficient system management. The Presentation Layer allows buyers, sellers, and administrators to access the platform through web or mobile browsers. The Business Layer defines the responsibilities of different users within the system. The Service Layer handles important functionalities such as user authentication, product and order management, NLP-based search, image-based search, and payment processing. The Data Layer is responsible for storing and retrieving information using technologies like MongoDB and ImageKit. User requests pass through these layers for processing and data handling before responses are returned to the users. This architecture improves scalability, maintainability, reliability, and overall system performance.

5.2 CONTEXT FLOW:-

5.2.1 DIAGRAM:-

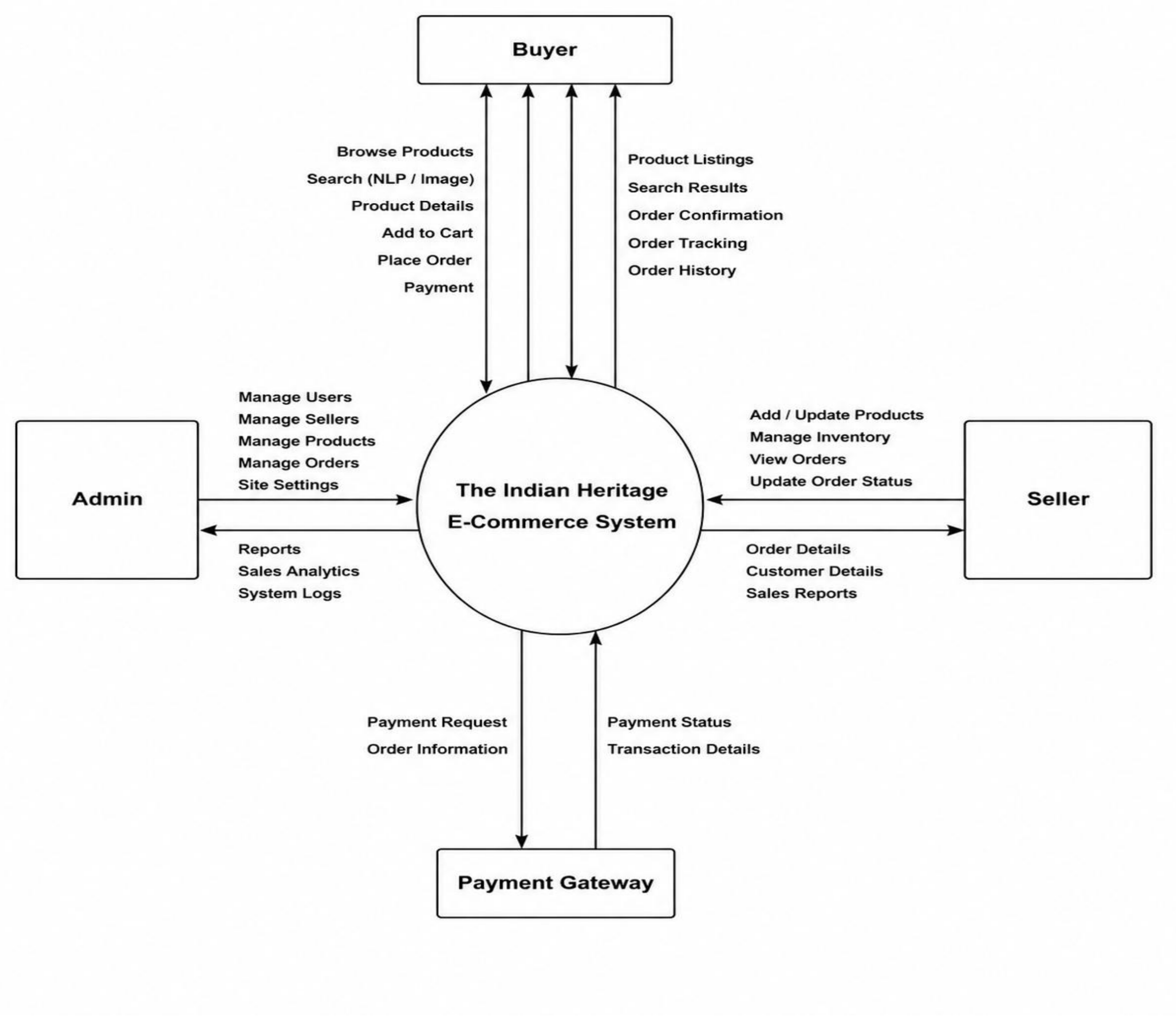

Figure 5.2.1 : Architecture Diagram

5.2.2 Explanation:-

The above-mentioned diagram shows Level 0 context flow of The Indian Heritage e-commerce system. This diagram clearly depicts the interactions of the central system with the external entities like the Buyer, Seller, Admin, and the Payment Gateway, along with their data flows. The center system can be termed as the system's processor that receives inputs from the user and gives out the respective output to them. The Buyer is one of the interacting parties who browse the available products, performs keyword, NLP-based, and Image-based search, views product details, makes orders and pays. It receives its outputs from the center in the form of product lists, results of searches performed, order confirmation, and other information related to the same. Seller on the other hand takes care of the available products on his end by adding/updating the products, managing their inventories, and fulfilling orders. He gets the orders and customer details from the center system. The Admin is responsible for taking care of the platform as a whole and he gets information from the system in terms of reports and analyses. Payment Gateway is another external party whose responsibility is to make transactions safe.

Detailed Design

CHAPTER 6

DETAILED DESIGN

6.1 USE CASE DIAGRAM:-

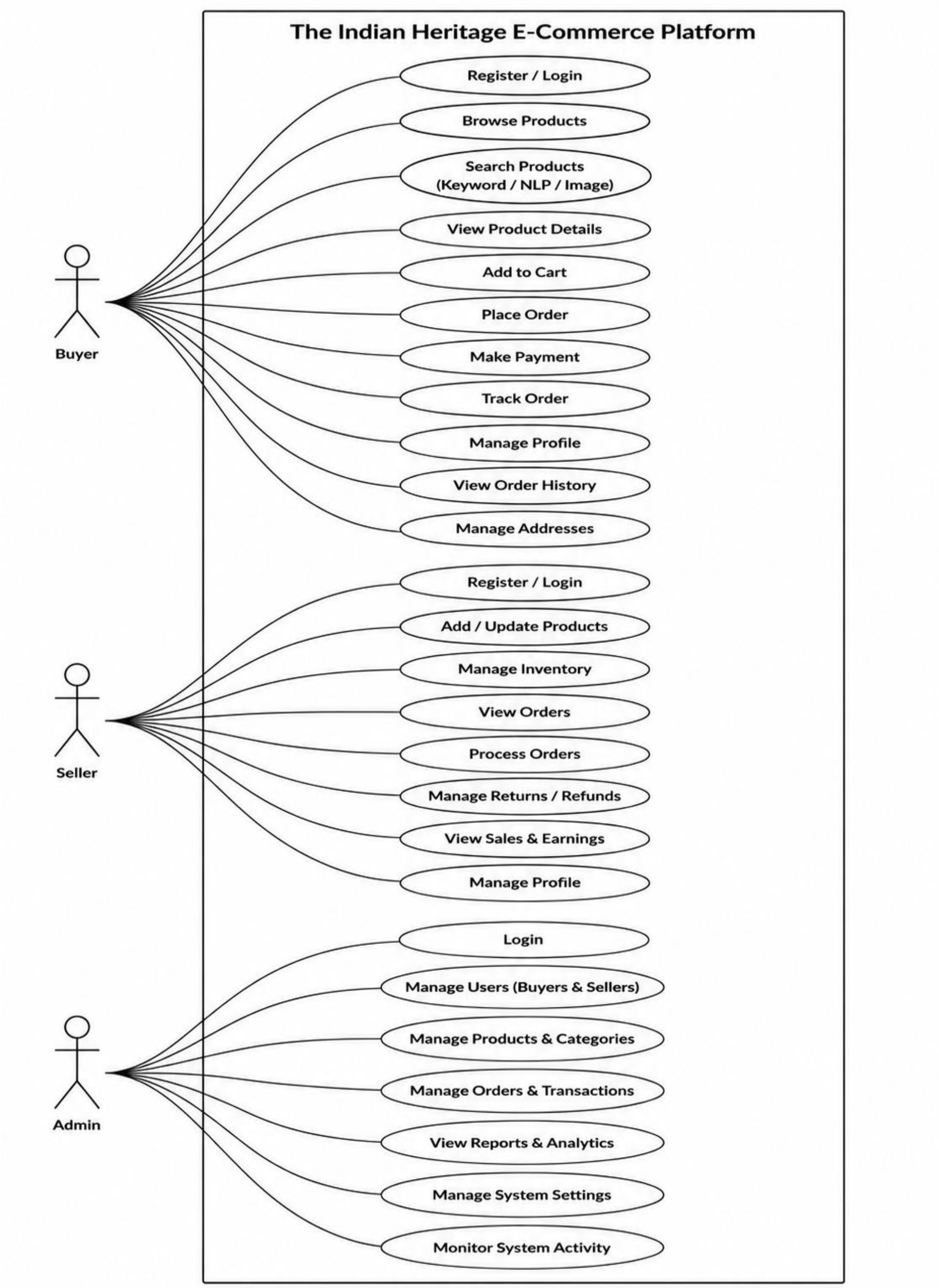

Figure 6.1.1 : Use Case Diagram

EXPLANATION:-The use case diagram demonstrates the interface among various actors and the features provided by The Indian Heritage e-commerce application. The boundaries of the system comprise all the essential features, whereas the actors, namely Buyer, Seller, and Admin, are engaged with the system externally. The Buyer acts as the prime user who performs actions such as registration or login, searching products by keyword search, natural language processing search, or image search, displaying product detail, adding products to the shopping cart, placing an order, making a payment, and tracking order history. The Seller works with the system by managing product listing activities, including product listing, updating product listings, and updating order status. Finally, the Admin manages all aspects of the platform, including managing users, tracking activities on the system, updating product data, and generating reports for analysis.

6.2 ACTIVITY DIAGRAM:-

6.2.1 USER REGISTRATION FLOW:-

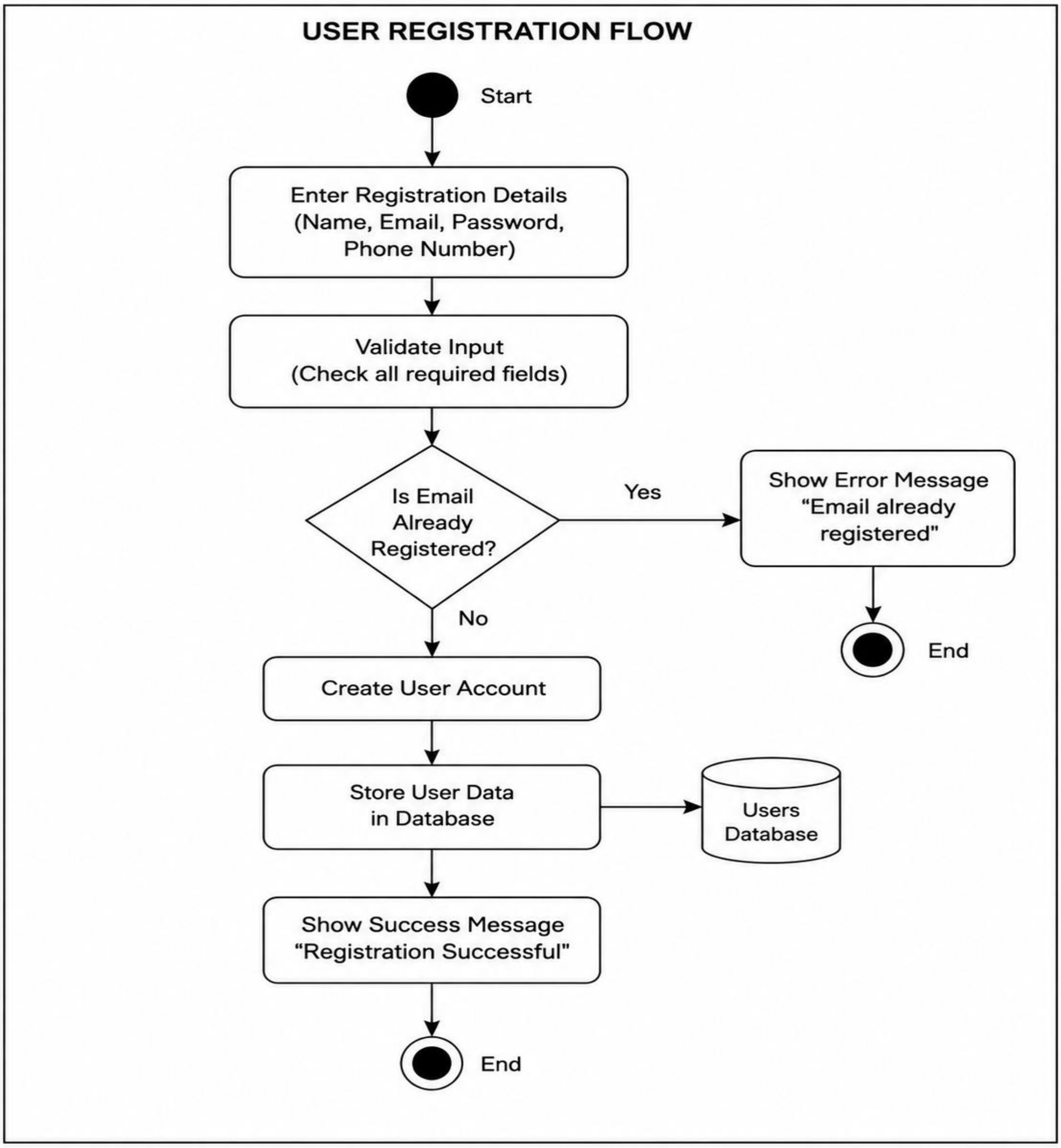

Figure 6.2.1 : USER REGISTRATION FLOW

EXPLANATION:-

The user registration procedure begins when a new user provides essential information such as name, email address, and password through the registration page. After submission, the system validates the entered information to ensure that all mandatory fields are completed correctly. The system then verifies whether the entered email address already exists in the database. If the email is already registered, an error message is displayed and the registration process is stopped. However, if the email address is unique, the system creates a new user account and securely stores the user details in the database. This process helps maintain secure and valid user registration within the platform.

6.2.2 LOGIN FLOW:-

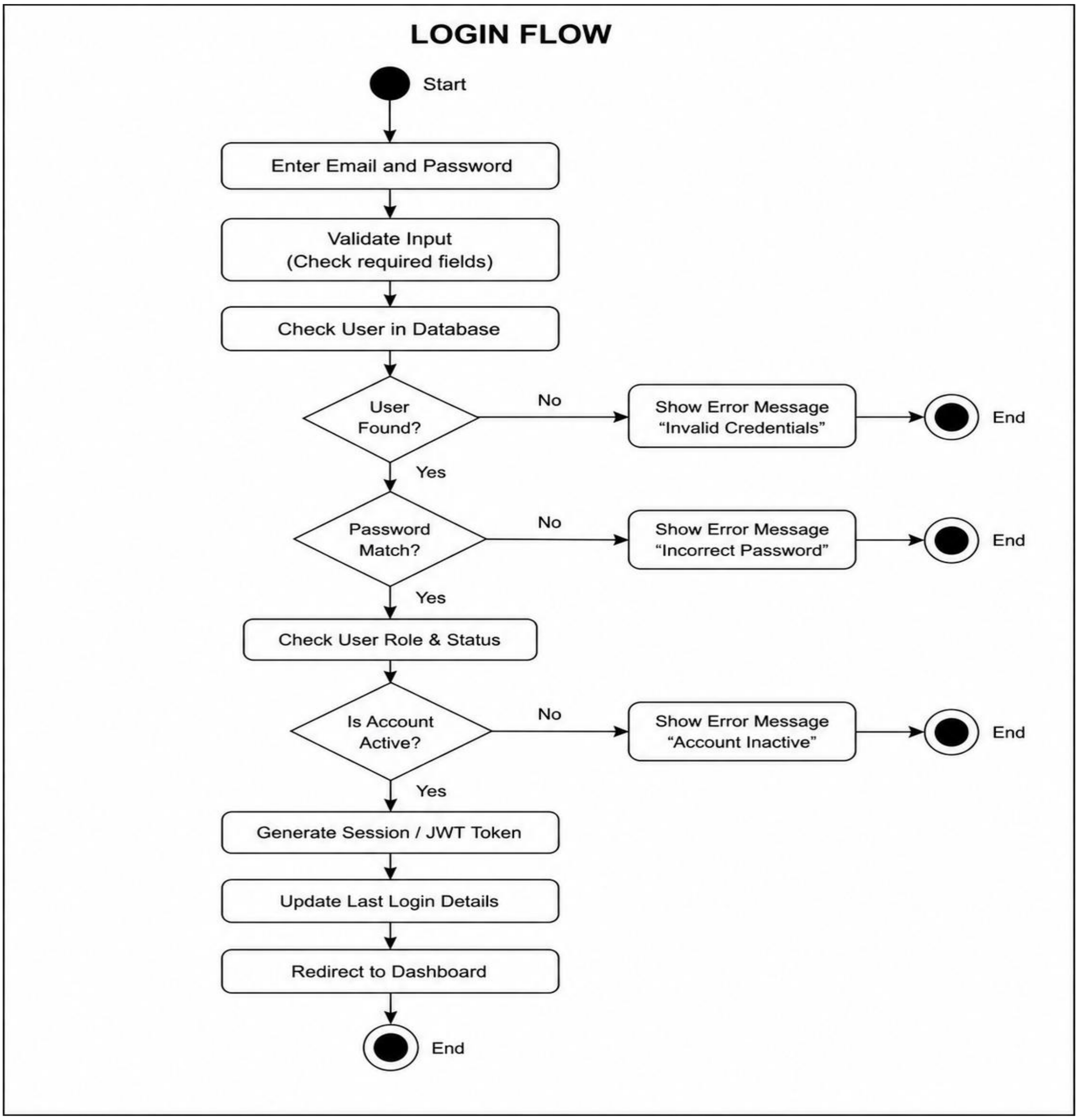

Figure 6.2.2 : LOGIN FLOW

EXPLANATION:-

The login process begins when a registered user enters their email address and password through the login interface. After submitting the credentials, the system performs validation to ensure that all required fields are properly filled. The system then verifies whether the user account exists in the database. If the account is not found, an error message indicating invalid login credentials is displayed and the process is terminated. If the user exists, the system proceeds to verify the entered password. In case the password does not match, the system displays an error message and stops the login process. If the password is correct, the system further checks whether the user account is active and authorized for access. Once all validations are completed successfully, the system generates a secure session or authentication token, stores the login details, and redirects the user to the dashboard page.

6.2.3 PRODUCT SEARCH & BROWSING FLOW:-

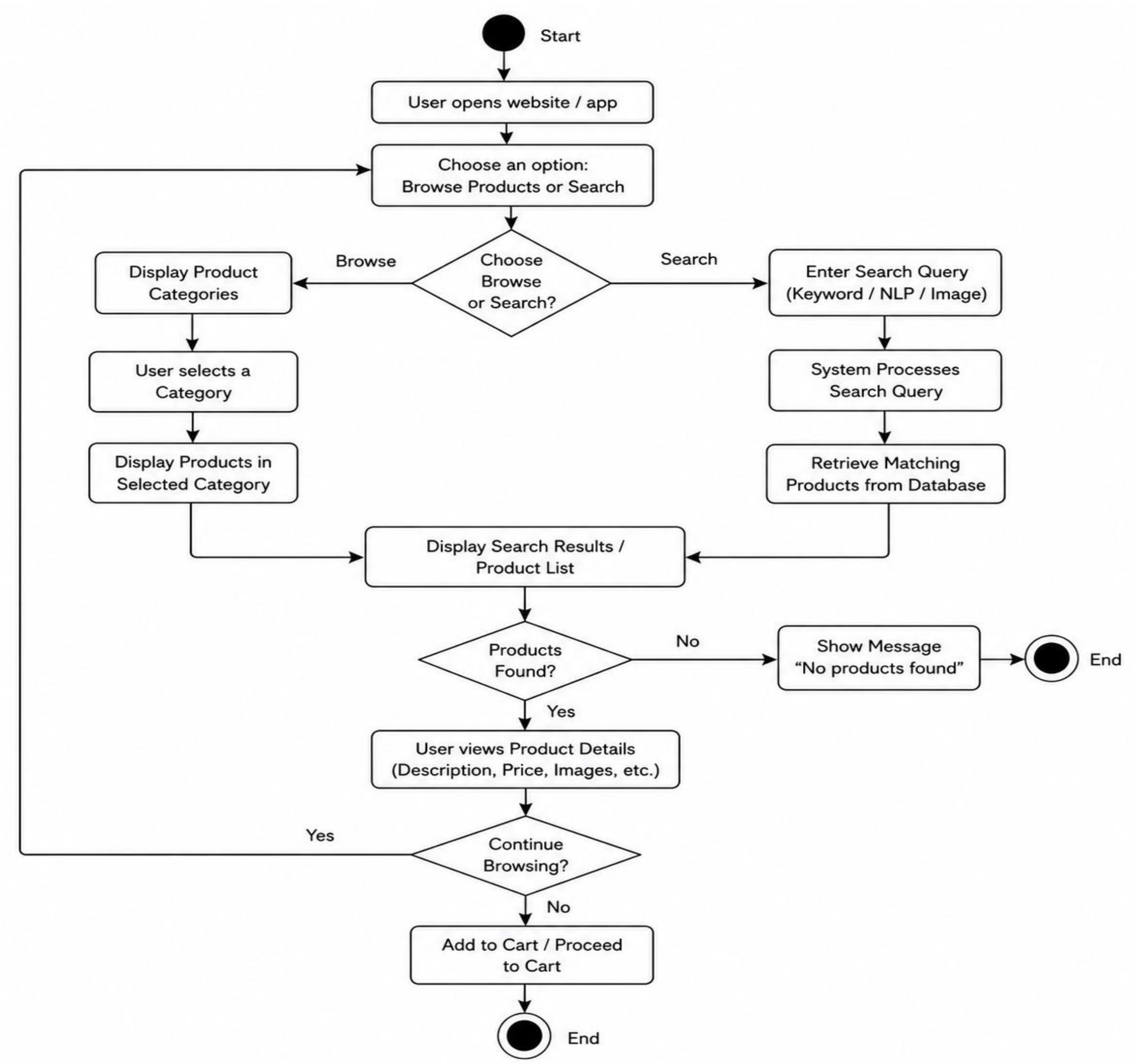

Figure 6.2.3 : PRODUCT SEARCH AND BROWSING FLOW

EXPLANATION:-

The product search and browsing process begins when the user accesses the application and chooses either to browse products or search for a specific item. If the user selects browsing, the system displays products under different categories for easier navigation. When the user performs a search, they can use keywords or advanced search methods such as NLP-based search or image-based search. The system processes the entered query and retrieves matching products from the database. If no related products are found, the system displays an appropriate message. However, if matching products are available, the user can view complete product details including description, price, and images before proceeding further.

6.2.4 ORDER PAYMENT FLOW:-

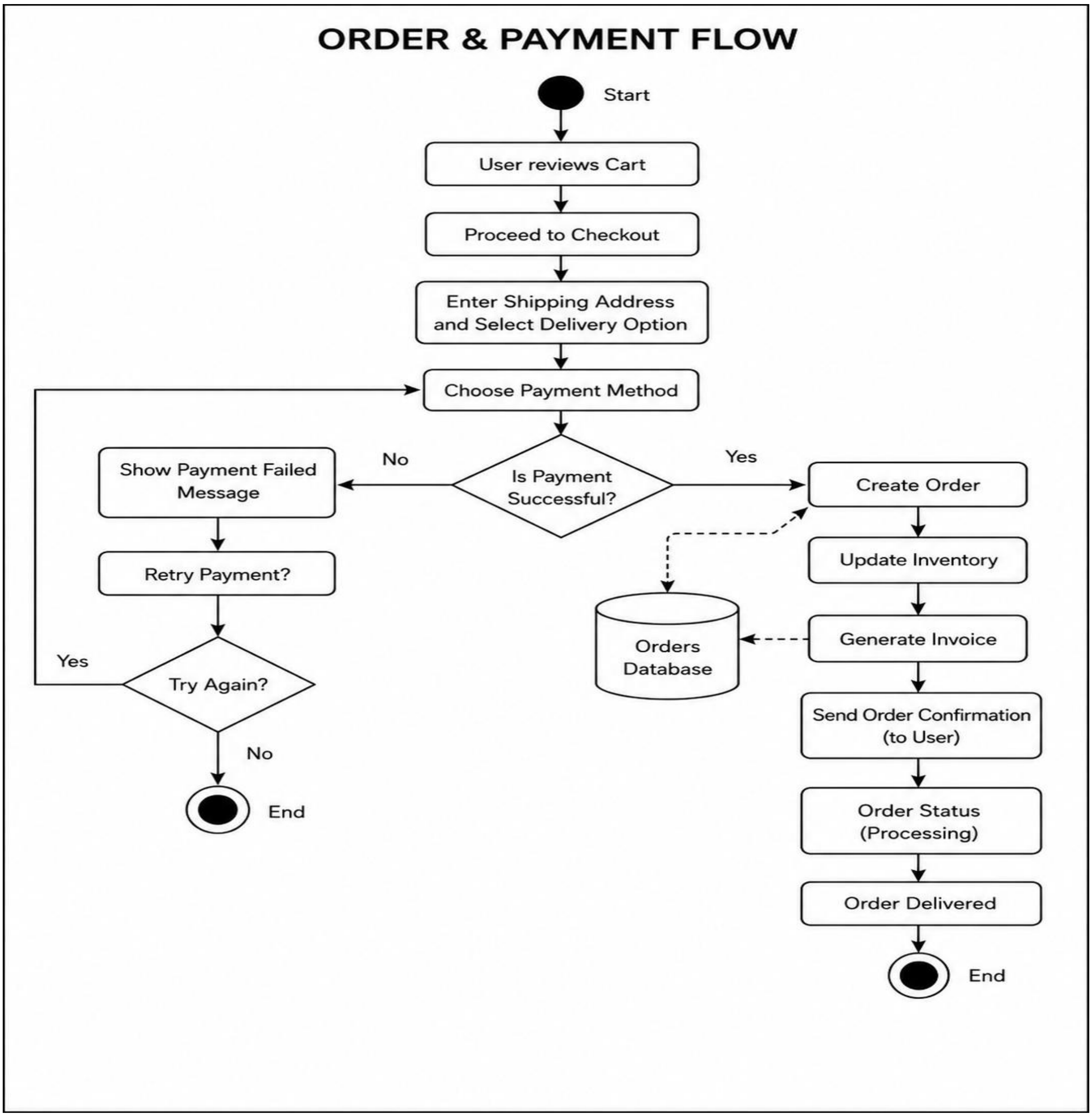

Figure 6.2.4 : ORDER PAYMENT FLOW

EXPLANATION:-

The order and payment process begins when the user reviews the products added to the cart and proceeds to checkout. The system then requests the user to enter shipping information and select a suitable delivery option. After confirming the details, the user chooses a payment method to complete the transaction. The payment request is then forwarded to the payment gateway for verification and processing. If the payment is unsuccessful, the system displays an error message and allows the user to retry the payment or cancel the process. However, if the payment is completed successfully, the system generates the order, stores the order information in the database, updates the product inventory, and creates an invoice. Finally, an order confirmation message is sent to the user, and the order status is updated accordingly.

6.2.5 SELLER PRODUCT MANAGEMENT FLOW:-

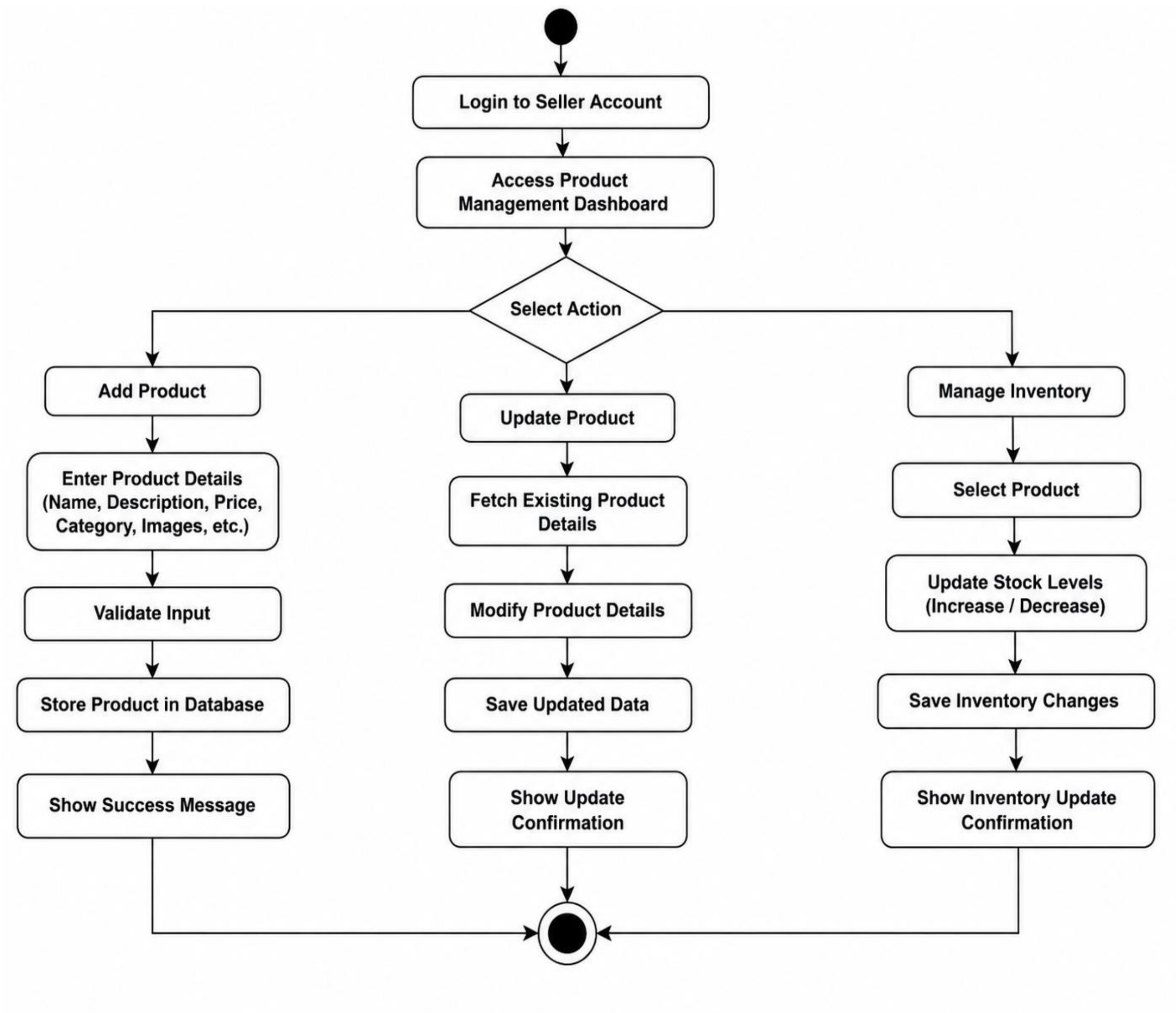

Figure 6.2.5 : SELLER PRODUCT MANAGEMENT FLOW

EXPLANATION:-

TThe seller product management process begins when the seller logs into the platform and accesses the product management section. The seller can add new products, update existing product information, or manage product inventory through the dashboard. While adding a product, details such as product name, description, category, price, and product images are entered into the system. The system validates the entered information before storing it securely in the database. If the seller decides to update product details, the system retrieves the existing information and allows necessary modifications before saving the changes. The seller can also adjust stock availability to maintain accurate inventory records. All updates made by the seller are reflected on the platform in real-time, ensuring that product information remains accurate and up to date.

6.2.6 ADMIN MANAGEMENT FLOW:-

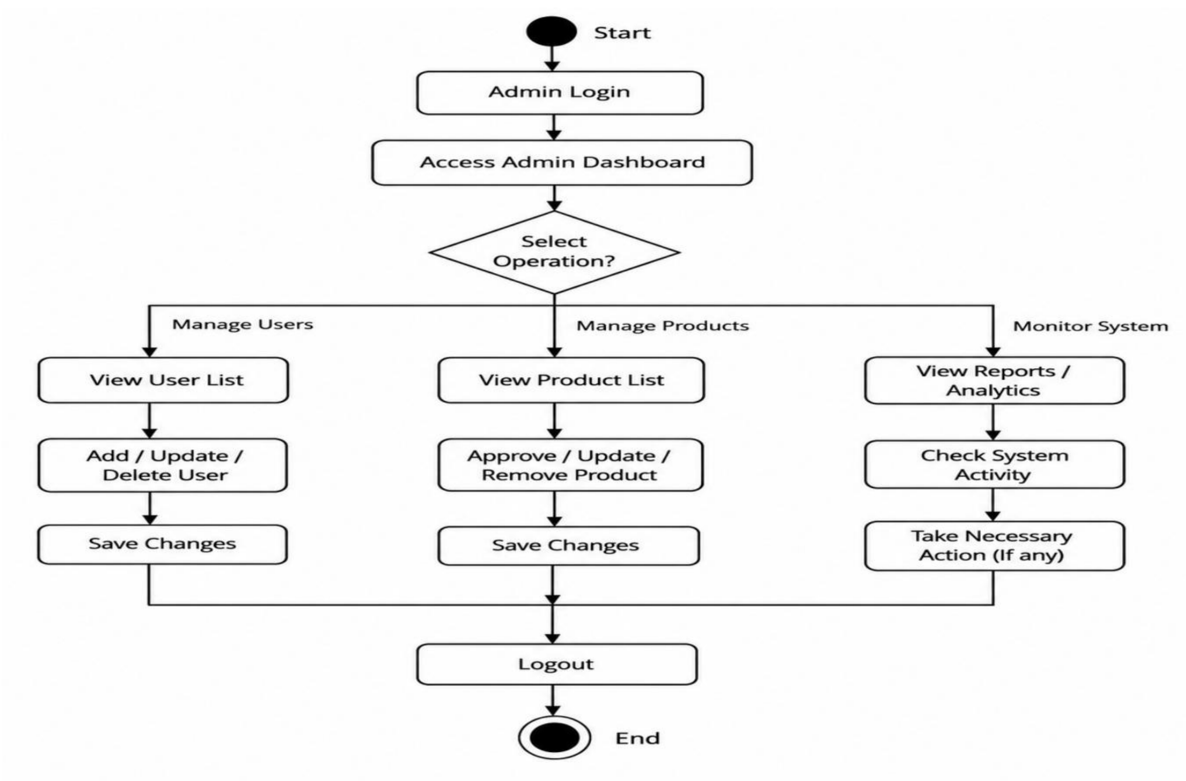

Figure 6.2.6 : ADMIN MANAGEMENT FLOW

EXPLANATION:-

The admin management process begins when the administrator logs into the system using valid credentials and accesses the admin dashboard. From the dashboard, the admin can perform various operations such as managing users, managing products, and monitoring overall system activities. In user management, the admin can view user details, update information, or remove users when necessary. In product management, the admin can review product listings, update product details, or remove inappropriate products from the platform. The system also provides monitoring and reporting features that help the admin track user activities and analyze overall platform performance. All modifications made by the administrator are stored and updated in the database to ensure proper system maintenance and smooth platform operation.

6.3 DATABASE DESIGN:-

6.3.1 INTRODUCTION:-

The MongoDB database technology is utilized for storing all the data involved in the project. The use of this particular database type was chosen mainly due to its flexibility in dealing with different types of data. Given the nature of the project, where there will be multiple users and objects (such as products and orders), it would be more practical to have a flexible NoSQL database. All data will be stored as documents in collections, which provides efficient storage and fast retrieval. This database will allow for the implementation of all the basic features provided by the platform. The following section describes some of the main collections used in the project.

6.3.2 COLLECTIONS:-

6.3.2.1 USER COLLECTION :-

```
model User {
  id             String          @id @default(auto()) @map("_id") @db.ObjectId
  name           String
  email          String          @unique
  password       String
  role           Role            @default(BUYER)
  products       Product[]
  cart           Cart[]
  wishlist       Wishlist[]
  createdAt      DateTime        @default(now())
  updatedAt      DateTime        @updatedAt
  addresses      OrderAddress[]
  orders         Order[]
  reviews        Review[]
  sellerRequests SellerRequest[]
}
```

Figure 6.3.2.1 : User Collection

The User collection holds all the information related to the users, such as their name, email, password, and role within the system. These users can be both buyers and sellers and even admins. Furthermore, it is connected with other aspects of the system, for instance, products, orders, cart items, wishlist, and reviews. Also, this collection holds information about account creation and modification times. As almost all operations in the system are performed by users, this collection is of great importance.

6.3.2.2 SELLER REQUEST COLLECTION :-

```
model SellerRequest {
  id        String        @id @default(auto()) @map("_id") @db.ObjectId
  userId    String        @db.ObjectId
  user      User          @relation(fields: [userId], references: [id])
  status    RequestStatus @default(PENDING)
  shopName  String?
  phone     String?
  address   String?
  gstin     String?
  createdAt DateTime      @default(now())
  updatedAt DateTime      @updatedAt
}
```

Figure 6.3.2.2 : Seller Request Collection

This list is created for users who wish to become sellers on the site. Rather than giving access to selling right away, the system will register a request which will include the seller's name, contact information, and other information. The status can be pending, approved, or rejected among others. This makes sure that only those people who are eligible to sell are granted permission by the admin.

6.3.2.3 PRODUCT COLLECTION :-

```
model Product {
  id                 String     @id @default(auto()) @map("_id") @db.ObjectId
  name               String
  description        String
  price              Float
  mainImage          String
  quantity           Int
  otherImages        String
  sizes              String[]
  colors             String[]
  state              String?
  categories         Category[]
  cart               Cart[]
  wishlist           Wishlist[]
  orders             Order[]
  reviews            Review[]
  userId             String     @db.ObjectId
  user               User       @relation(fields: [userId], references: [id])

  // Embedding tracking fields
  hasImageEmbedding  Boolean    @default(false)
  hasTextEmbedding   Boolean    @default(false)
  pineconeId         String?
}
```

Figure 6.3.2.3 : Product Collection

Product Collection consists of all products offered by the online marketplace. Information about each item, including its title, description, cost, picture, amount in stock, sizes, and colors, can be found here. Every product has its corresponding seller. The product information collection not only holds regular product data but also contains information for extended search functionality through embedding of pictures and texts.

6.3.2.4 CATEGORY COLLECTION :-

```
model Category {
  id        String  @id @default(auto()) @map("_id") @db.ObjectId
  name      String
  productId String  @db.ObjectId
  products  Product @relation(fields: [productId], references: [id])
}
```

Figure 6.3.2.4 : Category Collection

The Categories collection makes it possible to classify various products in separate categories. The categories are associated with certain products, thus allowing users to easily navigate among products depending on their categories. This way, users will not have to conduct searches for products all the time.

6.3.2.5 CART COLLECTION:-

```
model Cart {
  id        String  @id @default(auto()) @map("_id") @db.ObjectId
  userId    String  @db.ObjectId
  user      User    @relation(fields: [userId], references: [id])
  productId String  @db.ObjectId
  product   Product @relation(fields: [productId], references: [id])
  quantity  Int     @default(1)
  color     String?
  size      String?
}
```

Figure 6.3.2.5 : Cart Collection

The Cart class saves the products chosen by the user before they make a purchase. It saves information about the user, product chosen, quantity, and additional product characteristics such as size and color. This helps the user to change or review his choices before making the purchase.

6.3.2.6 WISHLIST COLLECTION:-

```
model Wishlist {
  id        String  @id @default(auto()) @map("_id") @db.ObjectId
  userId    String  @db.ObjectId
  user      User    @relation(fields: [userId], references: [id])
  productId String  @db.ObjectId
  product   Product @relation(fields: [productId], references: [id])
  createdAt DateTime @default(now())

  @@unique([userId, productId])
}
```

Figure 6.3.2.6 : Wishlist Collection

The Wishlist feature enables users to store products that they like for future reference. The Wishlist module keeps track of both the user and the product, as well as the date and time when the product was added to their list.

6.3.2.7 ORDER ADDRESS COLLECTION:-

```
model OrderAddress {
  id        String  @id @default(auto()) @map("_id") @db.ObjectId
  userId    String  @db.ObjectId
  user      User    @relation(fields: [userId], references: [id])
  firstName String
  lastName  String
  address   String
  city      String
  state     String
  zip       String
  order     Order[]
}
```

Figure 6.3.2.7 : Order Address Collection

This set contains the details about the address used for delivery by the customer while placing the order. These fields include the name of the person, the address, the city, the state, and the zip code. Each address entry has an association with a user who can use it again while placing other orders.

6.3.2.8 ORDER COLLECTION:-

```
model Order {
  id            String       @id @default(auto()) @map("_id") @db.ObjectId
  quantity      Int
  status        OrderStatus  @default(PENDING)
  paymentMethod String       @default("Razorpay")
  usersId       String       @db.ObjectId
  user          User         @relation(fields: [usersId], references: [id])
  createdAt     DateTime     @default(now())
  updatedAt     DateTime     @updatedAt
  Product       Product?     @relation(fields: [productId], references: [id])
  productId     String?      @db.ObjectId
  addressId     String       @db.ObjectId
  address       OrderAddress @relation(fields: [addressId], references: [id])
}
```

Figure 6.3.2.8 : Order Collection

The Order contains data of all orders placed on the website. Information such as order number, order status, mode of payment, user who places the order, item purchased, and delivery address will be recorded in this collection. The timestamps of each order can also be captured.

6.3.2.9 REVIEW COLLECTION:-

```
model Review {
  id        String   @id @default(auto()) @map("_id") @db.ObjectId
  rating    Int
  comment   String
  image     String?
  userId    String   @db.ObjectId
  user      User     @relation(fields: [userId], references: [id])
  productId String   @db.ObjectId
  product   Product  @relation(fields: [productId], references: [id])
  createdAt DateTime @default(now())
  updatedAt DateTime @updatedAt
}
```

Figure 6.3.2.9 : Review Collection

The Review entity will store information provided by users for products, including ratings, comments, and images where possible. A relationship must exist between Reviews and Users as well as Products because each review belongs to a particular product and user.

6.3.3 ENUMS:-

6.3.3.1 ORDER STATUS :-

```
enum OrderStatus {
  PENDING
  COMPLETED
  CANCELLED
}
```

Figure 6.3.3.1 : Order Status

The Order Status enum provides all possible statuses of the order in the system. Some examples of such status include “Pending”, “Completed” and “Cancelled”. This helps in monitoring and managing the order status in the system.

6.3.3.2 USER ROLE :-

```
enum Role {
  BUYER
  SELLER
  ADMIN
}
```

Figure 6.3.3.2 : User Role

Role enum is used to define various roles in the application like Buyer, Seller, and Admin. Every user role will have its own set of permissions and responsibilities within the application. This will help implement role-based access control and ensure that users do not perform activities outside their respective roles.

Implementation

Chapter 7

Implementation

7.1 PSEUDO CODE WITH DESCRIPTION:-

7.1.1 USER AUTHENTICATION :-

```
FUNCTION authenticate_user(email, password):
    user ← find_user_in_database(email)

    IF user EXISTS AND verify_password(password, user.password):
        RETURN user.role
    ELSE:
        RETURN "Invalid Credentials"
END FUNCTION
```

7.1.2 FETCH PRODUCTS :-

```
FUNCTION fetch_products(filters):
    products ← query_database(filters)

    IF products IS EMPTY:
        RETURN []
    ELSE:
        RETURN products
END FUNCTION
```

7.1.3 SEARCH PRODUCTS :-

```
FUNCTION search_products(query):
    processed_query ← preprocess_query(query)

    IF query_type == "TEXT":
        results ← search_using_keywords_or_NLP(processed_query)

    ELSE IF query_type == "IMAGE":
        image_features ← extract_image_features(query)
        results ← match_similar_products(image_features)

    RETURN results
END FUNCTION
```

7.1.4 ADD TO CART :-

```
FUNCTION add_to_cart(user_id, product_id, quantity):
   product ← get_product(product_id)

   IF product.quantity >= quantity:
      update_cart(user_id, product_id, quantity)
      RETURN "Added to Cart"
   ELSE:
      RETURN "Out of Stock"

END FUNCTION
```

7.1.5 PLACE ORDER :-

```
FUNCTION place_order(user_id, cart_items):
   total_amount ← calculate_total(cart_items)

   order_id ← create_order(user_id, cart_items, total_amount)

   RETURN order_id
END FUNCTION
```

7.1.6 PAYMENT PROCESSING :-

```
FUNCTION process_payment(order_id, payment_details):
   payment_status ← initiate_payment_gateway(payment_details)

   IF payment_status == "SUCCESS":
      update_order_status(order_id, "COMPLETED")
   ELSE:
      update_order_status(order_id, "FAILED")

   RETURN payment_status

END FUNCTION
```

7.1.7 ORDER TRACKING :-

```
FUNCTION track_order(order_id):
   order_details ← fetch_order(order_id)

   RETURN order_details.status

END FUNCTION
```

7.1.8 PRODUCT MANAGEMENT (SELLER) :-

```
FUNCTION manage_products(seller_id, action, product_data):
    IF action == "ADD":
        add_product_to_database(product_data)

    ELSE IF action == "UPDATE":
        update_product(product_data)

    ELSE IF action == "DELETE":
        delete_product(product_data.id)

    RETURN "Success"
END FUNCTION
```

7.1.9 ADMIN CONTROL :-

```
FUNCTION admin_operations(action, data):

    IF action == "MANAGE_USERS":
        update_user_status(data)

    ELSE IF action == "MANAGE_PRODUCTS":
        moderate_products(data)

    ELSE IF action == "VIEW_REPORTS":
        generate_reports()

   RETURN "Operation Completed"

END FUNCTION
```

7.1.10 COMPLETE SYSTEM FLOW :-

```
BEGIN

INPUT: user_request

--- 1. USER AUTHENTICATION ---

user_role ← authenticate_user(email, password)

--- 2. PRODUCT BROWSING / SEARCH ---

IF user selects "Browse":

   products ← fetch_products(filters)

IF user selects "Search":

   products ← search_products(query)

--- 3. CART MANAGEMENT ---

IF user adds product:

   add_to_cart(user_id, product_id, quantity)

--- 4. ORDER PLACEMENT ---

IF user proceeds to checkout:

   order_id ← place_order(user_id, cart_items)

--- 5. PAYMENT ---

payment_status ← process_payment(order_id, payment_details)

--- 6. ORDER TRACKING ---

status ← track_order(order_id)

--- 7. SELLER OPERATIONS ---

IF user_role == "SELLER":

   manage_products(seller_id, action, product_data)

--- 8. ADMIN OPERATIONS ---

IF user_role == "ADMIN":

   admin_operations(action, data)

--- 9. FINAL RESPONSE ---

DISPLAY results to user interface

END
```

7.2 SCREENSHOTS

7.2.1 SignUp Page

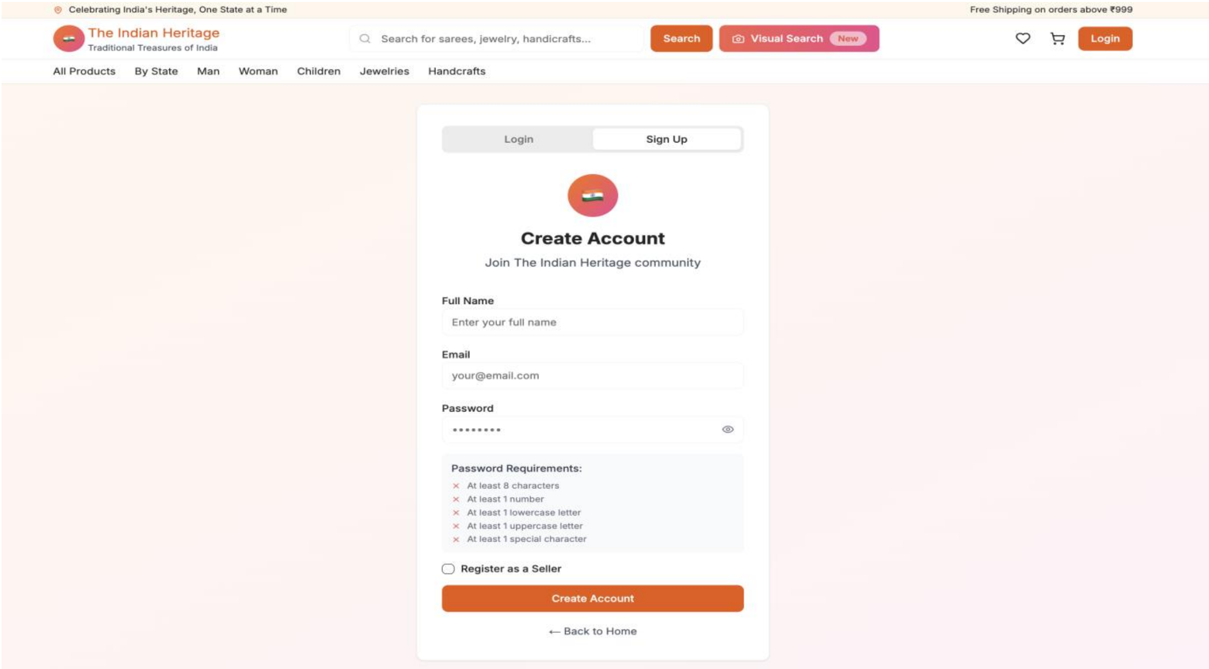

Figure: 7.2.1 Signup Page

This page depicts the user registration form where the user can register into the system. It asks for important credentials like name, email, and password from the user. It verifies whether the input data is correct or not, especially whether the email is unique or not. After the successful completion of registration, the user's account will be saved in the database.

7.2.2 Login page

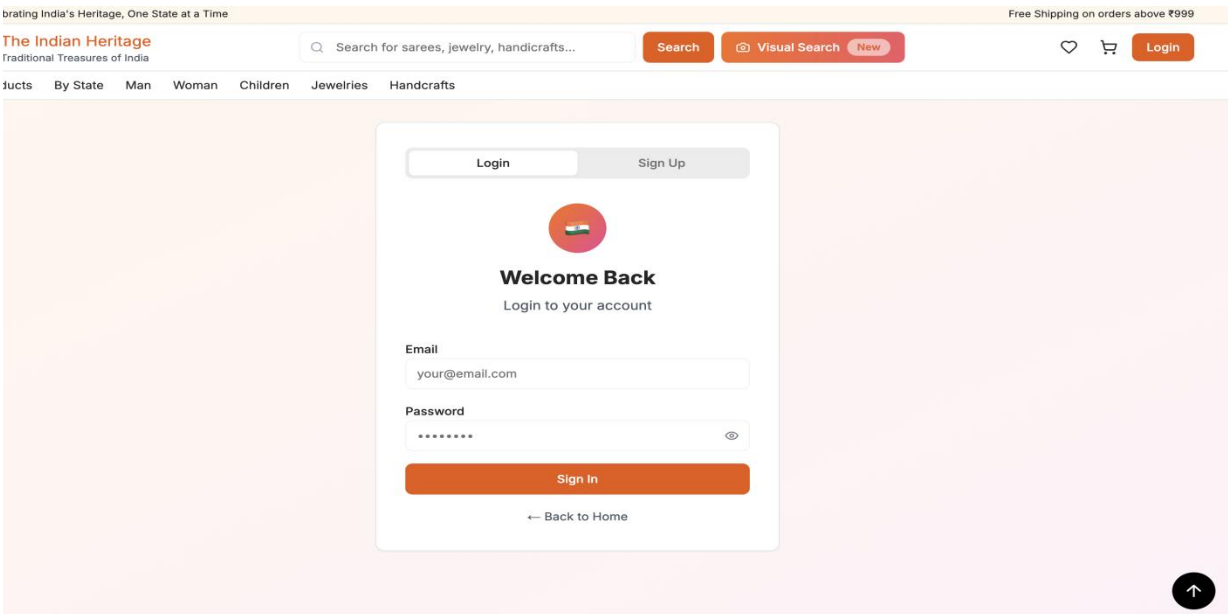

Figure: 7.2.2 Login Page

It is the login screen that allows registered users to gain access to their account by using their email address and password. The input values are then verified from the information stored in the database. In case of wrong entries, relevant messages appear. After successful login, the user will be directed to the main screen.

7.2.3 Home page

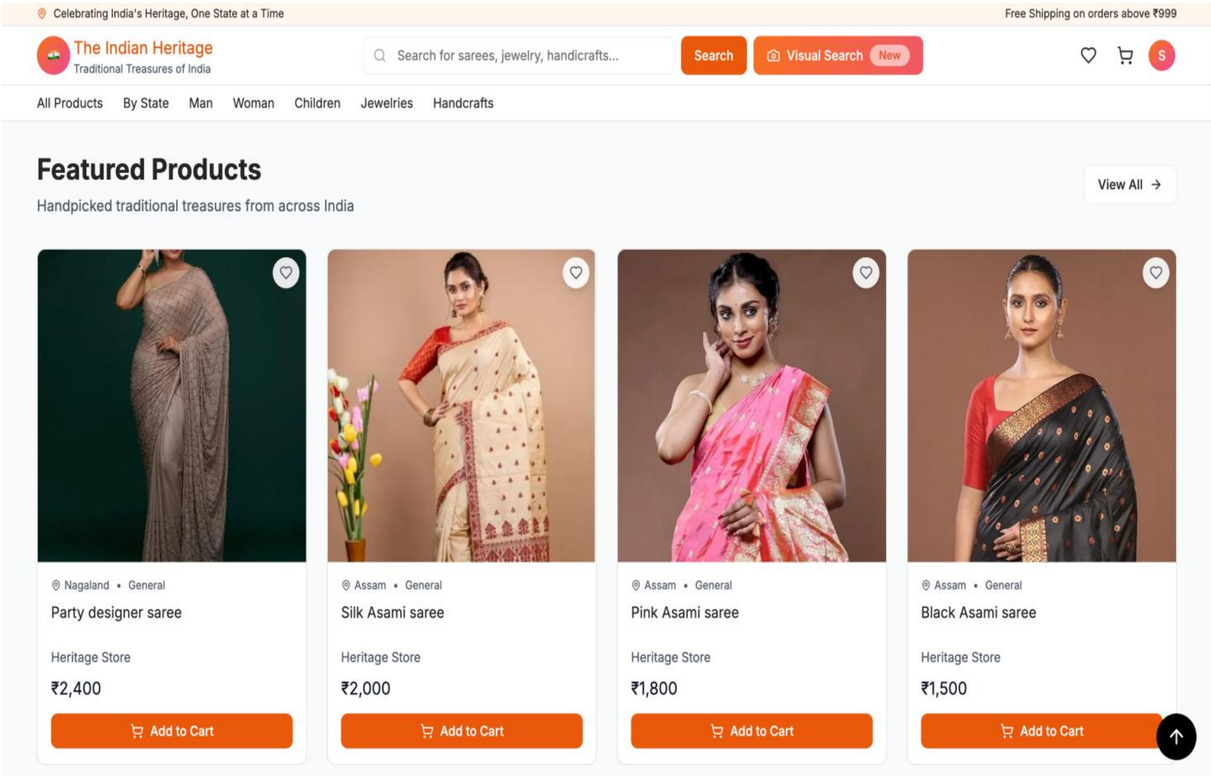

Figure: 7.2.3 Home page

Homepage acts as the landing page where users will be directed when accessing the site. This section gives information on the various products, product categories, and special featured products. Users can search products, make searches using the search bar, and also move around the site using the homepage. The homepage has been designed in such a way that it ensures easy navigation to important features like login, product categories, and the cart.

7.2.4 Product Listing Page

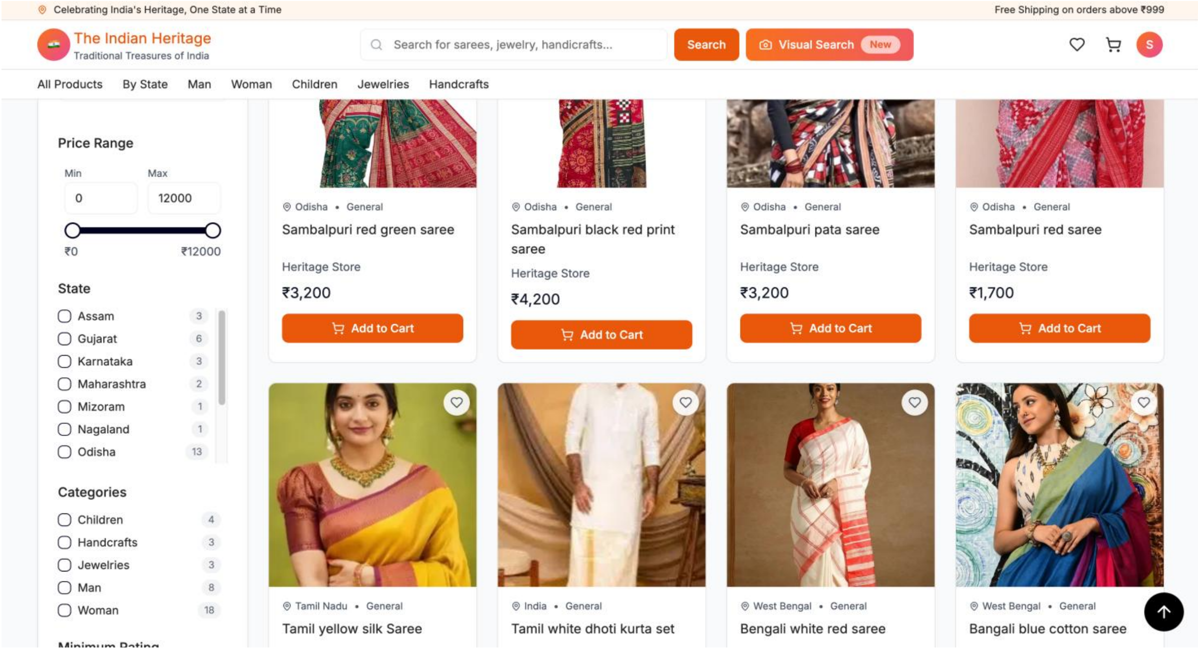

Figure 7.2.4 Product list page

This is the page where a number of products found in the platform are listed in an organized fashion. Customers can easily navigate through the available products using category listing or filtering according to criteria such as cost and type of product. Information about individual products includes images, product title and price which make it easier to compare various products. Searches can also be performed from this page.

7.2.5 Product Details Page

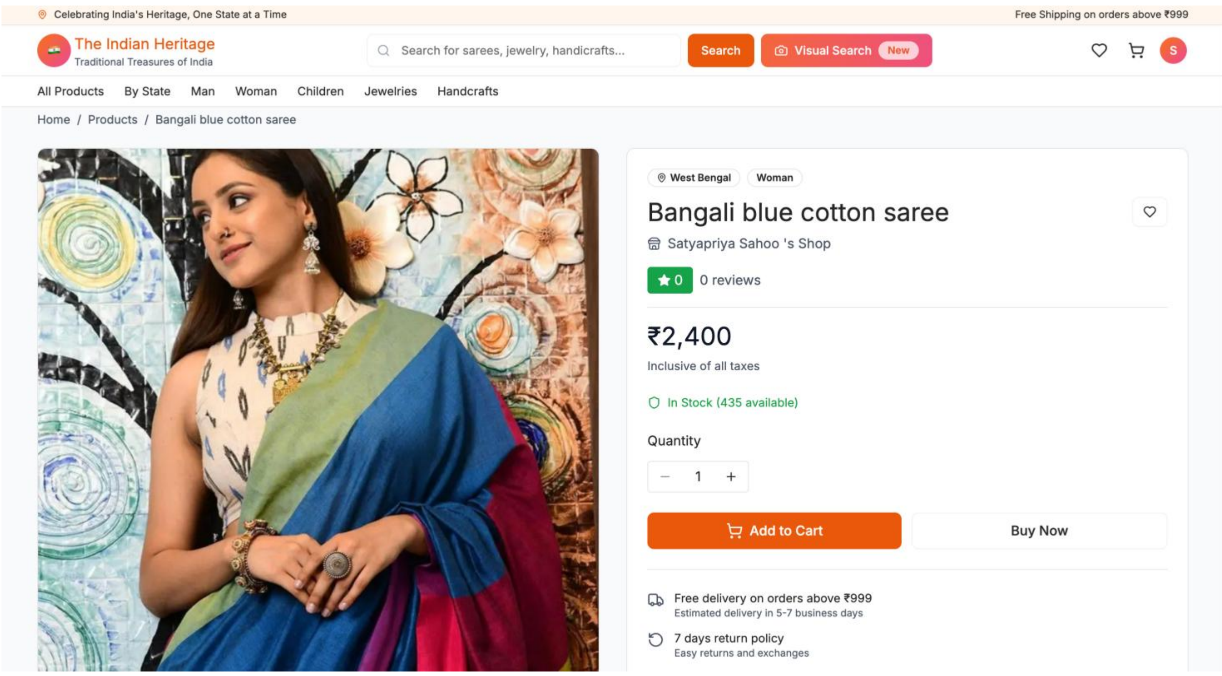

Figure 7.2.5 Product Details Page

This screen contains all the information related to the product chosen by the user. This will include data such as name of the product, price, size/ color, and stock. In addition, it gives an overview of all the information about that particular product. Users have the option to add this product to their shopping cart/wish list after they have made up their minds.

7.2.6 Cart Page

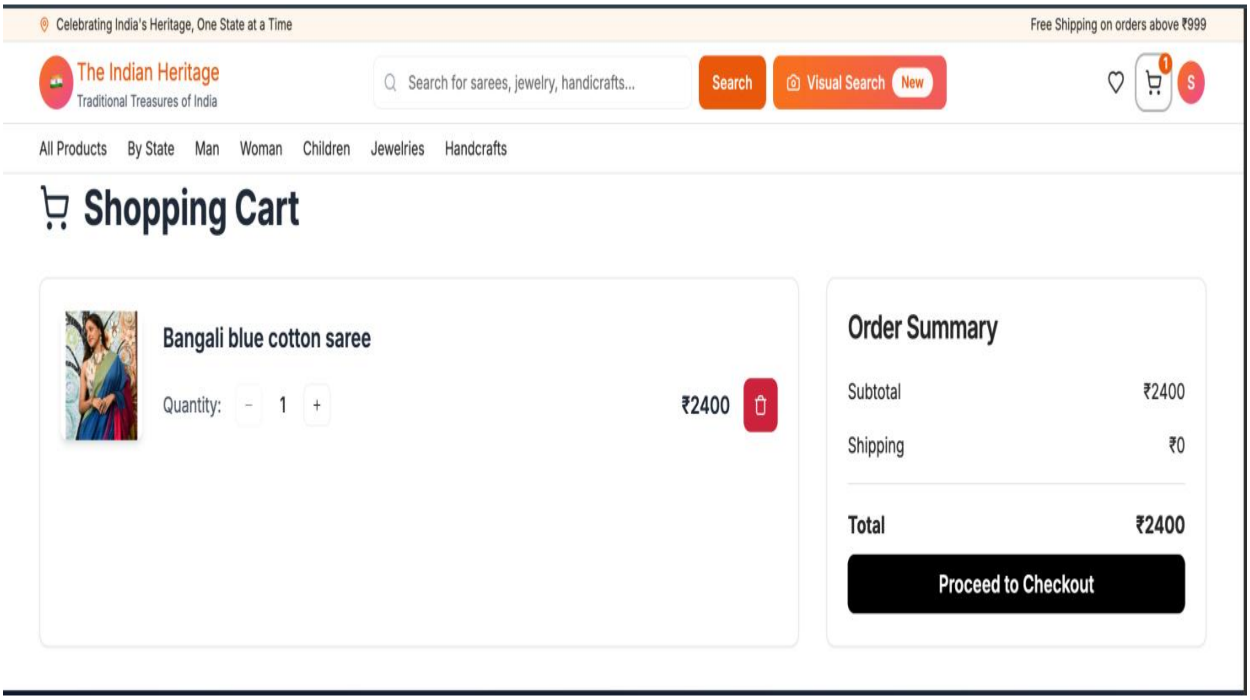

Figure: 7.2.6 Cart Page

This screen reflects those products chosen by the user to purchase. The screen reflects details like product name, image, quantity chosen by the user, price, and total price for all products in the cart. Users have the ability to change the quantity and remove products from their carts as required. Total price is calculated automatically after making any modification by the system.

7.2.7 Checkout/payment Page

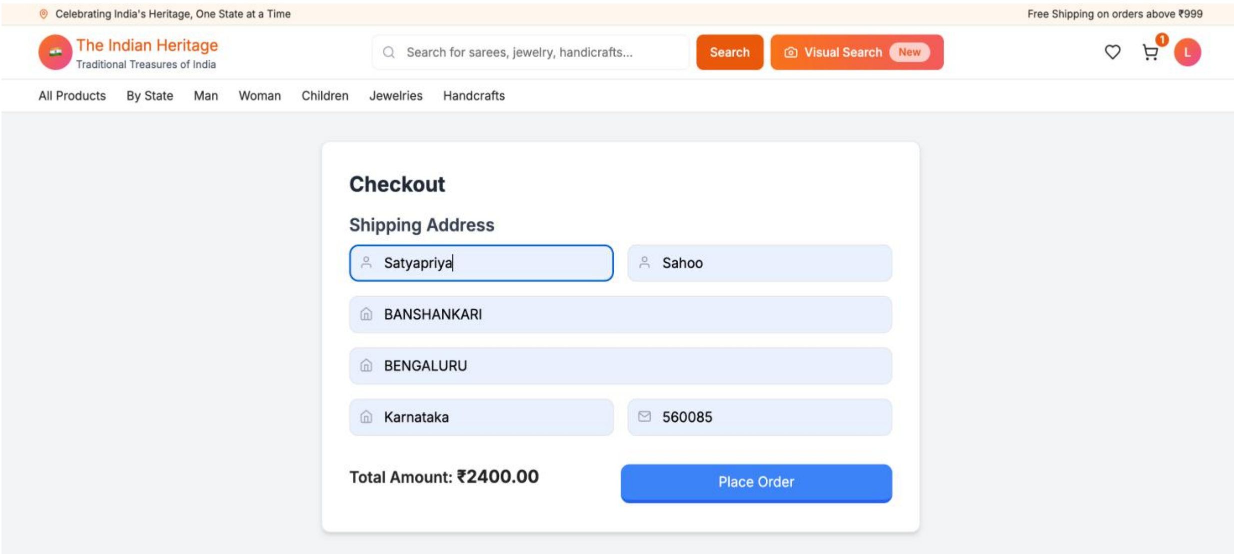

Figure: 7.2.7 Checkout/payment Page

This page is accessed to make the purchase after reviewing the cart contents. Users must provide their shipping address and opt for the preferred mode of payment. All relevant information pertaining to the order, including the total amount to be paid and order items, is provided on the page for verification. After initiating the payment process, the order request is processed via the payment gateway.

7.2.8 Order Conferm Page

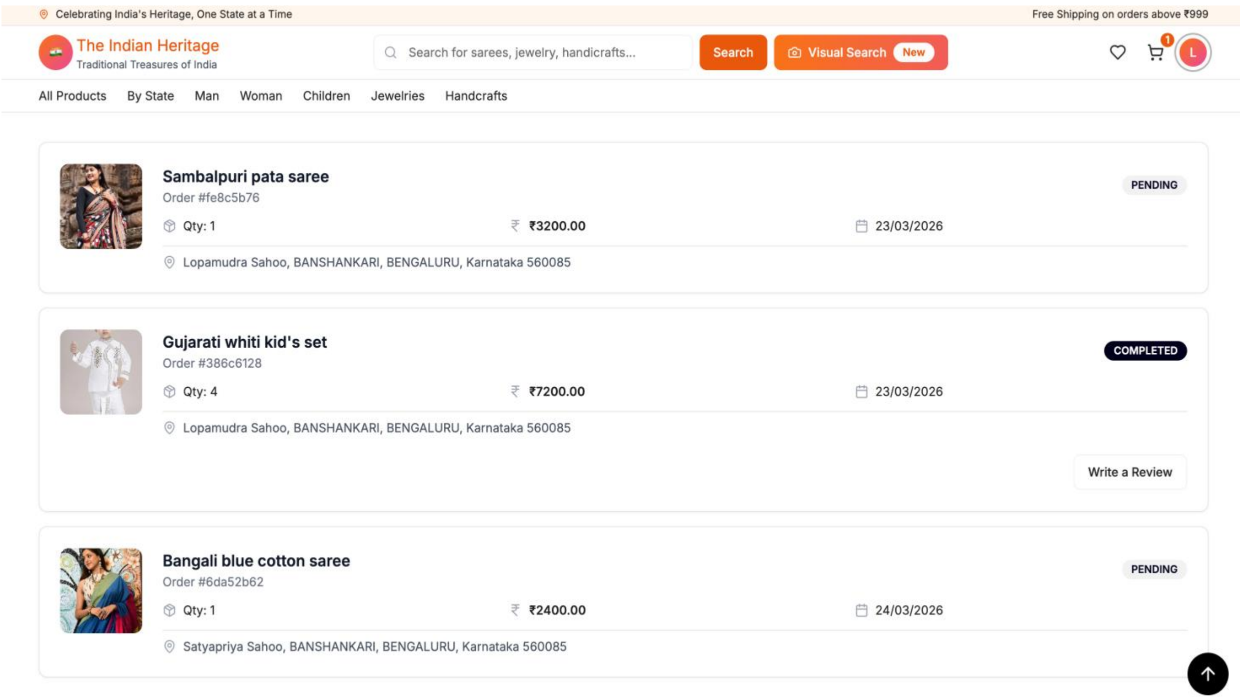

Figure: 7.2.8 Order Conferm Page

The order confirmation screen will appear when the customer successfully places an order. This screen includes the order confirmation details along with some other critical information like order ID, order content, and delivery details. The user gets an assurance that his/her order has been placed successfully.

7.2.9 Seller Dashboard

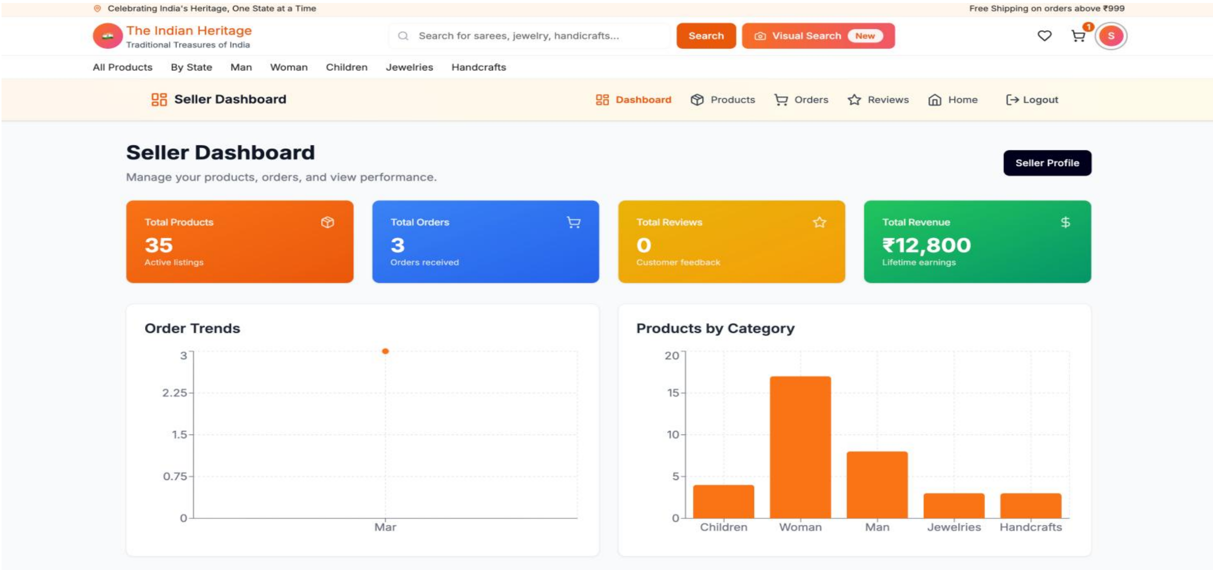

Figure: 7.2.9 Seller Dashboard

This is the seller's dashboard that enables them to manage their products and get relevant information about the orders placed on their products. It offers them an opportunity to create new products, edit their current listings, and check the inventory. In addition, they are able to see the orders of their products and know the status of the orders.

7.2.10 Product Management Page

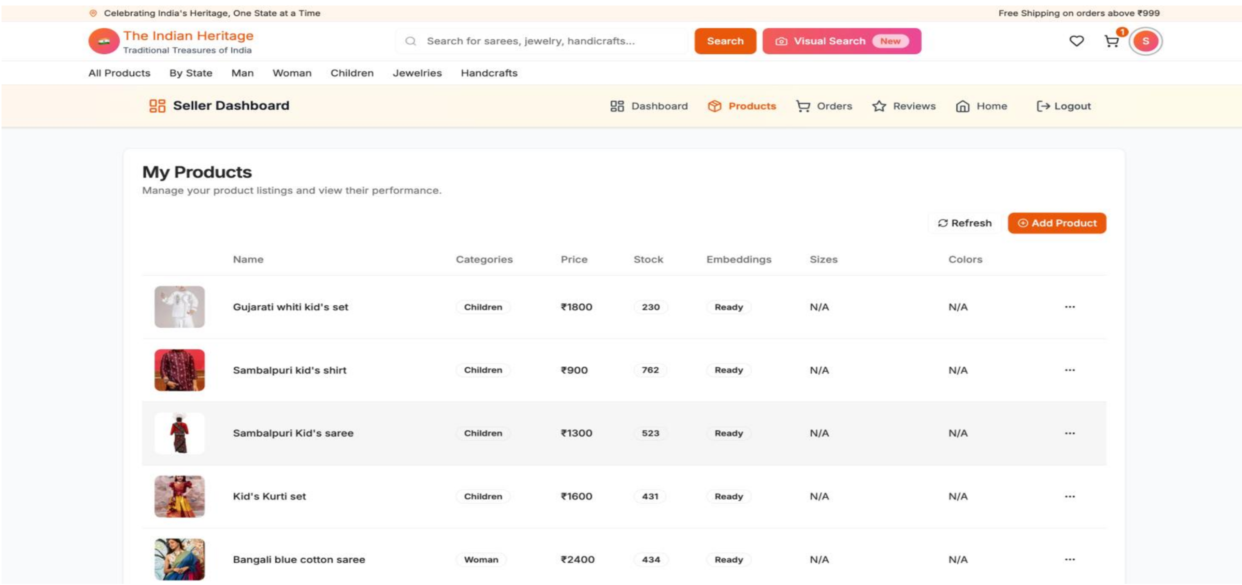

Figure: 7.2.10 Checkout/payment Page

The screen provides an option for sellers to have full management control of the products being sold. Sellers can create their own product items using details like the item's name, its description, its cost, photos of the product, and its current stock. Additionally, they can edit or remove their existing products depending on any changes in the stock levels. The software then validates the data input and displays the latest information instantly on the platform.

7.2.11 Admin Dashboard

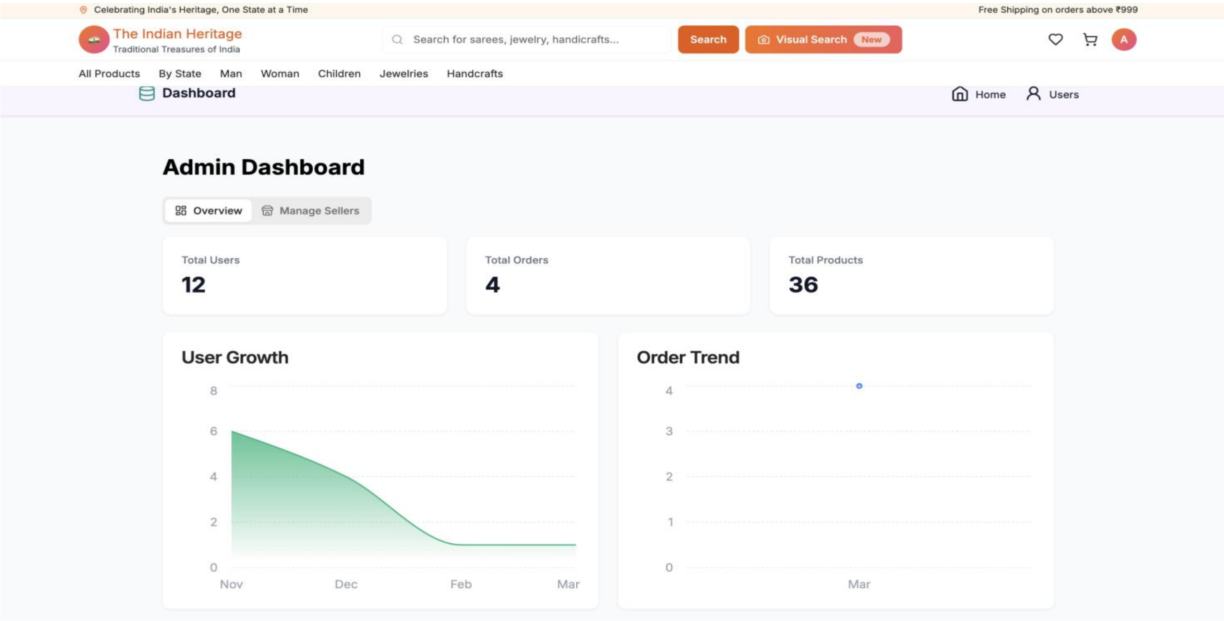

Figure: 7.2.11 Admin Dashboar

The above dashboard is the main screen that acts as the administrative control panel for the management of all the system components. This dashboard highlights the system activity including management of user accounts and products, orders and so forth. The administrator is able to execute tasks like accepting or deleting certain products and managing users’ account and system data.

Software Testing

CHAPTER 8

SOFTWARE TESTING

8.1 User Interface Testing :-

In User Interface (UI) Testing, the goal was to make sure that the system offers a consistent user experience in terms of user-friendliness throughout the various screens of the application. The objective here was to make sure that all user interface objects such as the buttons, forms, inputs and other navigation options worked effectively and easily understandable by the users.

A number of interfaces, for instance the sign up, log in, product listings, shopping cart, checkout and order placement pages were evaluated with regard to the accuracy of data entry and whether error messages appeared where necessary. The consistency of layout and alignment of UI objects was also ensured during testing. In addition to that, the user interface response across different screen sizes was also assessed.

The ease with which users navigated within different sections was also checked while making sure that each button performs the right action, e.g., when the add to cart button is clicked the item must be added to the cart and if the proceed to checkout button is clicked the process will proceed accordingly.

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
UI-01	Signup form input	Accept valid data and show success	As expected	Pass
UI-02	Login validation	Show error for invalid credentials	As expected	Pass
UI-03	Navigation flow	Smooth page transitions	As expected	Pass
UI-04	Product display	Products displayed correctly	As expected	Pass
UI-05	Responsive design	Proper display on different devices	As expected	Pass

Table 8.1.1 : User Interace Test

8.2 Manual Test Case:-

8.2.1 Scenario: Login Test

This scenario verifies if users can login successfully by providing correct user details and if the system is able to show proper error messages in case of incorrect user input.

Test case ID	Description	Expected result	Actual result	Pass/fail
TC_LOGIN_001	Provide correct username and password and click login	User dashboard is shown	Success	Pass
TC_LOGIN_002	Provide incorrect password and click login	Show error message as "Invalid credentials"	Success	Pass
TC_LOGIN_003	Leave login form blank and click login	Display validation error	Success	Pass

Table 8.2.1 : Login Test

8.2.2 Scenario for Testing: Registration Validation

The following test is conducted to validate whether new users can register or not and whether validation occurs on an appropriate basis.

Test Case ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_REG_001	Register using valid information	Registration success	Registration successful	Pass
TC_REG_002	Register using an invalid email Address	Error shown	Error displayed properly	Pass
TC_REG_003	Do not fill required fields	Validation error shown	Error shown properly	Pass

Table 8.2.2 : Registration Test

8.2.3 Scenario: Product Search & Listing

This test makes sure that the product search and listing function work well.

Test Case ID	Steps	Tested Results	Results Received	Pass/Fail
TC_PROD_001	Using keywords to search for products	Product listings returned	Returned results	Yes
TC_PROD_002	Searching for products using images	Product listings returned	Returned results	Yes
TC_PROD_003	Sorting products	Product listings sorted	Sorted results	Yes

Table 8.2.3 : Login Test

8.2.4 Scenario: Add to Cart

This is to check adding products to cart and dealing with stock scenarios.

Test Case ID	Steps	Expected Output	Actual Output	Pass/Fail
TC_CART_001	Add Valid Product to Cart	Product added successfully	Product added	Pass
TC_CART_002	Add Product out of stock	Display Out of Stock error	Correctly displayed error	Pass

Table 8.2.4 : Add To Cart Test

8.2.5 Scenario: Order and Payment

This test covers the entire order and payment procedure.

Test Case ID	Steps	Expected Output	Actual Output	Passed/Failed
TC_ORDER_001	Carry out ordering using correct information	Successful order placement	Order placed successfully	Passed
TC_ORDER_002	Successfully complete payment process	Payment confirmed successfully	Payment completed successfully	Passed
TC_ORDER_003	Test case for payment failure	Display payment error	Payment error displayed correctly	Passed

Table 8.2.5 : Order Payment Test

8.2.6 Scenario: Seller Product Management

This test verifies the ability of the sellers to manage their products correctly.

Test Case ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_SELL_001	Create new product	Product created successfully	Product is available	Pass
TC_SELL_002	Edit product information	Information updated successfully	Product updated	Pass
TC_SELL_003	Delete product	Product deleted	Product deleted	Pass

Table 8.2.6 : Seller Product Management Test

8.2.7 Scenario: Admin Operation

It tests the operations related to admin features like management of users and products.

Test Case ID	Operations	Expected Output	Actual Output	Status
TC_ADMIN_001	User Management	Operation performed	Operation successful	Passed
TC_ADMIN_002	Product Management	Operation performed	Operation successful	Passed
TC_ADMIN_003	Reporting	Reports generated	Reports loaded successfully	Passed

Table 8.2.7 : Admin Operations Test

Conclusion

CHAPTER 9

CONCLUSION

Indian Heritage is the name of the e-commerce web application which has been created as a user-friendly tool for effective searching and purchasing the desired products. This program successfully combines the most necessary elements, such as customer registration, searching for certain goods, adding items to the shopping cart, placing orders, as well as making safe payments. Alongside these main elements, there are other advanced components integrated in the application, including NLP-based search and image-based product search. The design of this application involves modern technologies and is well-structured, which makes its work flawless and allows managing the web application easily. There are different types of users involved in this system, including buyers, sellers, and administrators. With appropriate testing and validation, it can be said that the performance of this software application is flawless.

Future Enhancements

CHAPTER 10

FUTURE ENHANCEMENT

The Indian Heritage e-commerce platform can be enhanced further by introducing additional advanced features that improve both user experience and overall system efficiency. Personalized recommendation systems based on user interests, browsing behavior, and purchase history can help users discover products more effectively.

The search functionality can also be improved by enhancing NLP-based search and image-based search accuracy for better product discovery. More advanced filtering options such as price range, product ratings, availability, and location can make browsing easier and more convenient. Features like seller performance analytics, product review analysis, real-time order tracking, and notification systems can also improve platform usability.

From a technical perspective, implementing a microservices-based architecture and optimizing database operations can improve scalability and performance. Additionally, support for multiple languages and regional content can expand the platform’s accessibility. In the future, integrating AI-based analytics and intelligent recommendation systems can make the platform more efficient, scalable, and user-centric.

Appendix A :References

Appendix A

BIBLIOGRAPHY

I. B. Mitra and N. Craswell, "Neural Information Retrieval for E-Commerce," arXiv preprint arXiv:1705.01509, 2018.
This work improves semantic search techniques for better product retrieval and enhances intelligent product discovery in e-commerce systems.

II. ACM SIGIR Workshop, "Improving Product Search with NLP Techniques," Proc. ACM SIGIR Workshop, 2020.
This study enhances search accuracy using NLP, improving query understanding and delivering more relevant product search results.

III. S. Author et al., "Semantic Search Using Deep Learning," IEEE, 2021.
This paper uses deep learning to enable semantic search, allowing systems to understand context and provide accurate results.

IV. J. Johnson, A. Karpathy, and L. Fei-Fei, "Deep Learning for Image-Based Product Search," arXiv preprint arXiv:1603.09320, 2016.
This research enables image-based product search using deep learning, allowing users to find products through visual inputs.

V. S. Author et al., "Visual Search in E-Commerce Using CNNs," IEEE, 2020.
This work applies CNNs for visual search, improving product discovery through efficient image matching techniques.

VI. S. Author et al., "Design of Scalable E-Commerce Systems," IEEE, 2018.
This paper focuses on scalable system design to handle large user traffic and ensure efficient performance in e-commerce platforms.

VII. N. Dragoni, S. Dustdar, S. Larsen, and M. Mazzara, "Microservices vs Monolithic Architecture," arXiv preprint arXiv:1704.00154, 2017.
This study compares architectural approaches and helps in selecting efficient system design strategies for web applications.

VIII. S. Kaur and R. K. Gupta, "Security Issues in Online Payment Systems," 2019.
This work highlights security challenges in online payments and provides methods to ensure secure transaction processing.

IX. P. R. Singh and A. Mehta, "E-Commerce Growth in India," 2020.
This paper discusses the expansion of e-commerce in India and supports the relevance of modern online shopping platforms.

X. S. Author et al., "Digital Payment Adoption in India," IEEE, 2020.
This study explains the rise of digital payments and their role in enabling secure and efficient e-commerce transactions.

Appendix B :Plagiarism Report

Appendix B : References

PLAGARISM REPORT

Appendix C :Poster

Appendix C

POSTER